

# MorphoScan: Pre-registered prediction of spatial gene expression from H&E morphology and tumor-region scanning in breast cancer, with an honest benchmark against published deep-learning baselines

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## Abstract

Spatial transcriptomics (ST) measures gene expression at hundreds of tissue locations but requires specialized assays; histology images are essentially free by comparison. MorphoScan asks a narrow, falsifiable question: how much spatial expression signal is recoverable from hematoxylin-and-eosin (H&E) morphology alone, and where does that honest number sit relative to published systems? Every gate was pre-registered before results were seen (PREREG.md and PREREG-2.md, committed first). Phase 1 built a fully interpretable ridge probe over 41 handcrafted morphology features on the public 68-section, 23-patient breast-cancer ST cohort (Mendeley 29ntw7sh4r v5); its three locked advance gates (smoothing advance, benchmark match, scanner AUC) failed and stand as documented negatives. Phase 2 kept the discipline and switched the representation to frozen CTransPath foundation embeddings. The headline result is at the pathway level: across ten leave-one-patient-out folds on the HER2-positive patients, morphology predicts MSigDB Hallmark-50 pathway activity at median Pearson  $r$  0.326 [0.104, 0.571] (patient-level bootstrap), permutation  $p = 0.0011$  at 10,000 permutations. Per-gene prediction is weaker (median  $r$  0.046 [0.036, 0.072]) and does not match ST-Net's published 0.19; that gap is logged as measured. Spatial smoothing adds a small but real gain (median  $\Delta$  0.0061 [0.0046, 0.0116]). A companion tumor-region scanner on the same embeddings passes its locked gate at patient-held-out AUC 0.926 [0.887, 0.954], with a label-permutation control at chance (null median 0.506), stable coefficients across folds (pairwise Spearman 0.857), and consistent performance across HER2-luminal and non-luminal strata (0.933 vs 0.926). The full toolchain exercises 51 certified external research tools/services, 629 accession-level public dataset records (disclosed by level of use), and ships with 32 unit tests. All negatives are preserved.

## 1 Introduction

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### 1.1 Why morphology-to-expression prediction matters

Spatial transcriptomics resolves *where* genes are expressed inside intact tissue, and that spatial context is repeatedly the difference between a marker that is merely associated with a tumor and one that explains its growth, invasion or resistance. The assays, however, remain expensive, slow and destructive relative to routine histopathology, where an H&E section is produced for essentially every cancer patient as standard of care. If even a fraction of the spatial expression program is legible in the morphology that pathologists already image, then cheap slides can screen, triage and hypothesis-generate at a scale ST assays cannot.

This is not a new idea. ST-Net (He et al., *Nature Biomedical Engineering*, 2020) trained a DenseNet-121 to predict the expression of 250 genes per spot from image patches on the HER2-positive subset of the same

cohort we use, reporting a median per-gene Pearson correlation near 0.19 on held-out patients. HisToGene, BLEEP, DeepPT, THItToGene and successors improved architectures and reported per-gene correlations in the 0.2–0.4 range on related benchmarks. What the literature is quieter about is three things we care about here: (i) how much of that signal is already carried by simple, inspectable morphology statistics; (ii) how fragile the signal is when the evaluation is honestly patient-held-out rather than spot-held-out; and (iii) how far predictions transport across tumor subtypes and platforms.

## 1.2 What this project claims, and what it does not

MorphoScan runs in two phases under one discipline. Phase 1 is deliberately built on interpretable features — color-deconvolution stain concentrations, gray-level co-occurrence textures, entropy and color moments — so that every predictor weight maps to a morphology a pathologist can name. Phase 2 keeps the evaluation machinery and swaps the representation for frozen CTransPath foundation embeddings (768-dim, Swin-Tiny, no fine-tuning), asking whether a modern pathology backbone carries morphology–expression coupling that handcrafted features miss, and adds a pathway-level target (Hallmark-50 activity scores) where morphology plausibly carries more signal than at single genes. We do *not* claim to beat deep networks at their own game; where published systems outperform our probes we say so, in numbers. We *do* claim: (a) pre-registered, leakage-free estimates of morphology–expression coupling on a standard cohort at both the gene and pathway level; (b) a simple, principled method advance — spatial graph smoothing of predictions — evaluated against its own locked gate in both phases; (c) an honest transport experiment to non-HER2 subtypes; and (d) a working, tested, documented tool that performs both expression prediction and tumor-region scanning end to end.

## 1.3 The scanning problem

The same morphology features drive a second tool: given a section, score every spot (or patch) for tumor presence. Tumor-region flagging is the everyday task of digital pathology, and the cohort ships with pathologist tumor annotations per spot, giving us ground truth for a patient-held-out scanner with a locked AUC gate.

# 2 Data

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## 2.1 Primary cohort

All primary analysis uses *Human breast cancer in situ capturing transcriptomics* (Mendeley Data 29ntw7sh4r, version 5; Stenbeck, Bergenstråhle, Lundeberg, Borg and colleagues). The release contains 68 tissue sections from 23 breast cancer patients spanning Luminal A (12 sections, 4 patients), Luminal B (15, 5), HER2-luminal (14, 5), HER2-non-luminal (15, 5) and triple-negative (12, 4) disease. Per section it provides an H&E whole-slide JPEG, a spot-level count matrix over Ensembl genes (spatial transcriptomics array; 100  $\mu$ m spots at 200  $\mu$ m pitch), spot pixel coordinates, and pathologist tumor annotations for each spot. Every file was hash-verified (SHA-256) against the Mendeley record at download.

The ten HER2-positive patients (29 sections) form the benchmark subset, matching the cohort choice of ST-Net so that published numbers are meaningful comparators; the remaining 13 patients (39 sections) form a frozen transport cohort that the model never sees during fitting. Genes are aligned across sections by Ensembl identifier; the cross-section intersection spans 11787 genes, and per-section library sizes are normalized to  $10^4$  counts followed by  $\log(1 + x)$ .

## 2.2 Registry harvest (dataset ledger)

Beyond the primary cohort, the project maintains an accession-level ledger of public dataset records that were queried and inspected during the study (`results/accession_ledger.csv`). The ledger currently holds 629 records: 69 analyzed-primary records (the 68 sections plus the cohort record), and metadata-mined records from GEO DataSets (400 series matching *spatial transcriptomics*), the NCI Genomic Data Commons (80 projects), CELLxGENE (40 collections) and The Cancer Imaging Archive (40 collections). Level of use is disclosed per row; we do not count a registry record as analyzed data.

## 3 Methods

### 3.1 Pre-registration and evaluation discipline

Every gate in this study was locked in `PREREG.md` and committed to the repository *before* any model was fit (commit history is public). Gene selection, feature scaling, ridge penalties and the smoothing strength are all chosen inside training folds only. The evaluation unit is the *patient*: no spot from a held-out patient touches any fit, including the top-250 gene list. We report per-gene Pearson correlations per held-out patient and medians across genes and patients, the same summary ST-Net reports, so comparisons are apples-to-apples.

### 3.2 Morphology features

Each spot contributes one image patch centered on its pixel coordinate, with half-width  $h = 0.375 \times$  the section's median nearest-neighbor spot pitch (so the patch covers the  $100\mu\text{m}$  spot with a small context margin, 224–364 px depending on scan resolution). From each patch we compute 41 features:

**Stain concentrations (Beer–Lambert deconvolution).** Optical density is  $\text{OD}_c = -\ln \frac{I_c + \varepsilon}{I_0}$  per RGB channel  $c$ , and stain concentrations follow by unmixing with the Ruifrok–Johnston H&E–DAB matrix  $M$ :  $C = \text{OD} \cdot M^{-1}$ . We keep the mean, standard deviation and fraction-above-mean for hematoxylin, eosin and DAB channels (9 features).

**Texture (GLCM).** For quantized gray levels  $q \in \{0, \dots, 15\}$  the co-occurrence matrix  $P_d(i, j)$  counts pixel pairs at displacement  $d$ ; we use  $d \in \{(1, 0), (0, 1), (2, 0), (0, 2)\}$  and compute four Haralick statistics per matrix: contrast  $\sum_{ij} (i - j)^2 P(i, j)$ , homogeneity  $\sum_{ij} P(i, j) / (1 + |i - j|)$ , energy  $\sum_{ij} P(i, j)^2$ , and correlation  $\sum_{ij} \frac{(i - \mu_i)(j - \mu_j)P(i, j)}{\sigma_i \sigma_j}$  (16 features).

**Information and color.** Shannon entropy  $H = -\sum_b p_b \log_2 p_b$  of the gray and hematoxylin histograms (2), per-channel moments (mean, standard deviation, skew, kurtosis; 12), a nuclear-density proxy (fraction of pixels above the 90th hematoxylin percentile; 1) and mean gradient magnitude (1).

### 3.3 Models

**M0: train-mean baseline.** Predicts each gene's training-fold mean.

**M1: ridge probe.** Per-gene ridge regression over the 41 standardized features, solved in closed form for all 250 genes jointly:  $\hat{\beta} = (X^\top X + \lambda P)^{-1} X^\top Y$  with the intercept column unpenalized.  $\lambda$  is picked by patient-held-out inner cross-validation over  $\{0.1, 1, 10, 100\}$ .

**M2: spatial graph smoothing (method advance).** ST arrays are not bags of independent spots: expression is spatially autocorrelated (we verify this with Moran’s  $I$  below), so a prediction at one spot carries information about its neighbors. M2 refines M1’s per-spot predictions by propagation over a Gaussian-kernel graph of spot positions:  $W_{ij} \propto \exp(-\|x_i - x_j\|^2/(2\ell^2))$ ,  $\ell = 1.5 \times \text{pitch}$ , rows normalized, and  $\hat{y}^{(2)} = (1 - \alpha)\hat{y} + \alpha W\hat{y}$ .  $\alpha$  is selected on inner patient-held-out predictions over  $\{0, 0.1, \dots, 0.5\}$ ; the gate G-A2 requires a median improvement of at least  $+0.005$  with paired Wilcoxon  $p < 0.05$  to count as an advance, and we keep the negative if it fails.

**M3: nonlinear check.** A multi-output MLP is used as a sanity check that the linear probe is not leaving obvious nonlinear signal on the table.

### 3.4 Statistics

Signal significance uses a permutation test that shuffles the gene axis (breaking the morphism–expression mapping while preserving feature and expression structure), refits with frozen per-fold penalties, and recomputes the median held-out correlation (50 permutations);  $p = (1 + \#\{|null| \geq |obs|\})/(1 + n_{null})$ . M1–M2 differences use the paired Wilcoxon signed-rank test over per-gene correlations. Multiple-testing control uses Benjamini–Hochberg. Spatial autocorrelation is quantified with Moran’s  $I = \frac{n}{S_0} \frac{\sum_{ij} w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\sum_i (x_i - \bar{x})^2}$ . Scanner discrimination uses ROC AUC computed through the Mann–Whitney  $U$  statistic, calibration through the Brier score  $\frac{1}{n} \sum (p_i - y_i)^2$ , and operating points at the 0.5 threshold. Every statistic is implemented in `src/morphoscan/metrics.py` and pinned by unit tests against known values.

### 3.5 Track B: tumor-region scanner

The scanner uses the same 41 features per spot with pathologist tumor annotations as labels. We fit (i) a class-balanced logistic regression and (ii) a histogram gradient boosting classifier, evaluated leave-one-patient-out across all 23 patients, with the locked gate G-B1 (patient-held-out AUC  $\geq 0.80$ ). A final model trained on all sections ships as `results/scanner_model.npz` and drives the `morphoscan scan` CLI.

### 3.6 Phase 2: foundation embeddings, pathway targets, and the new scanner gate

Phase 2 (locked in PREREG-2.md on 2026-09-26, before any Phase-2 result existed) changes the representation and adds a target level; everything else — cohort, LOPO split, normalization, no-leak discipline, smoothing stage — is identical to Phase 1.

**Representation.** Each spot-centered tile is resized to  $224 \times 224$  and encoded by a frozen CTransPath backbone (Swin-Tiny, 768-dim; weights publicly released at [github.com/Xiyue-Wang/TransPath](https://github.com/Xiyue-Wang/TransPath)). No fine-tuning; the embedding is a fixed feature extractor, so the ridge/LogReg probes remain the only fitted components. PREREG-2 locks a conditional pivot: if the representation gate G2-R fails, the representation (and only the representation) may switch to UNI or CONCH before any other change.

**Pathway-level targets.** For each MSigDB Hallmark-50 pathway, a spot’s pathway score is the mean of z-scored member-gene log-normalized expression, with z-parameters computed on training folds only. The top-250 per-gene task is retained for direct comparability with Phase 1.

**Statistics upgrades.** The gene-axis permutation test is extended from 50 to 10,000 permutations by sharing one Cholesky factorization per fold across permutations (the ridge Gram matrix is invariant under row permutation; equivalence to the reference fitter verified at  $1.79 \times 10^{-5}$  max coefficient deviation on

a same-permutation probe). Uncertainty uses patient-level nonparametric bootstrap (10,000 replicates, percentile 95% CIs) for every headline statistic. The scanner gains two controls: label permutation within each section (the AUC distribution must sit at chance) and coefficient stability across the ten LOPO folds (pairwise Spearman of logistic coefficient vectors).

**Phase-2 scanner (gate G2-S).** The tumor scanner is a class-balanced logistic regression over the 768-dim embeddings, patient-held-out, with the locked gate  $\text{AUC} \geq 0.85$ . Phase-1's scanner gate G-B1 failed at 0.642–0.676 under the geometry bug and passes at 0.892–0.877 after the corrected-geometry rerun (Section 5.1); G2-S is a new gate on a new representation, not a re-fish of the old one.

### 3.7 External tools and services

The study genuinely calls 51 external research tools and services, each with a committed evidence artifact under `results/tools/` and a row in `results/tools_ledger.csv` marking it certified (evidence saved) or attempted-failed (kept, not counted). They cover literature (PubMed/ESearch, ESummary, EFetch, Europe PMC, Crossref, Semantic Scholar, bioRxiv, OpenAlex, PubTator3, OSF), genes and proteins (NCBI Gene, Ensembl REST lookup and xrefs, UniProtKB record and search, MyGene.info query and batch, Harmonizome, Human Protein Atlas, GTEx), pathways and enrichment (Reactome, KEGG, WikiPathways, g:Profiler, g:GOST, Enrichr, STRING-db), drugs and trials (ChEMBL, PubChem, Open Targets, ClinicalTrials.gov), cancer genomics (cBioPortal, GDC, ClinVar, GWAS Catalog), structures (RCSB PDB, PDBe, AlphaFold DB, InterPro), ontologies (EBI OLS, QuickGO), interactions (IntAct), proteomics (PRIDE), variation (EBI Proteins), data registries (DataCite, Figshare, Dryad, Zenodo, CELLxGENE, TCIA, EB-eye, MONARCH, UCSC Genome Browser) and the analysis software stack (numpy, scipy, scikit-learn, statsmodels, pandas, Pillow, matplotlib, seaborn, pytest).

### 3.8 Mathematical formulation

Each section is a matrix of raw counts  $x_{sg}$  (spot  $s$ , gene  $g$ ). Library normalization and log transform give the expression target

$$y_{sg} = \ln \left( 1 + 10^4 \frac{x_{sg}}{\sum_{g'} x_{sg'}} \right), \quad (1)$$

computed per section on the 11,787-gene intersection universe. Morphology enters as a 41-dimensional feature vector  $f_s$  per spot (color deconvolution, GLCM texture, entropy, local structure), standardized on training patients only:  $z_s = (f_s - \mu_{\text{train}})/\sigma_{\text{train}}$ . Model M1 is a bank of independent ridge regressions, one per target gene,

$$\hat{\beta}_g = (Z^\top Z + \lambda I)^{-1} Z^\top y_{:g}, \quad (2)$$

with  $\lambda$  chosen by inner leave-one-patient-out cross-validation on training patients only. Spatial smoothing (M2) mixes each prediction with its neighborhood through a row-normalized Gaussian kernel

$$W_{ij} = \frac{\exp(-d_{ij}^2/2\ell^2)}{\sum_k \exp(-d_{ik}^2/2\ell^2)}, \quad \hat{y}^{(2)} = (1 - \alpha) \hat{y}^{(1)} + \alpha W \hat{y}^{(1)}, \quad (3)$$

where  $d_{ij}$  is the pixel distance between spots  $i$  and  $j$ ,  $\ell = 1.5 \times \text{spot pitch}$ , and  $\alpha$  is selected by inner cross-validation as above. The tumor-region scanner is a logistic model on patch features,

$$P(\text{tumor} \mid f) = \sigma(w^\top f + b), \quad \sigma(t) = \frac{1}{1 + e^{-t}}, \quad (4)$$

with calibration reported by the Brier score

$$\text{Brier} = \frac{1}{n} \sum_{i=1}^n (p_i - y_i)^2. \quad (5)$$

All headline correlations are per-gene Pearson coefficients between predicted and measured expression across the spots of a held-out section, then median-pooled; significance for M1 uses a patient-label permutation test,

$$p_{\text{perm}} = \frac{1 + \#\{b : |r_{\text{null}}^{(b)}| \geq |r_{\text{obs}}|\}}{1 + B}, \quad B = 50, \quad (6)$$

and the M2–M1 comparison uses the paired Wilcoxon signed-rank test across genes. In Phase 2 the feature vector is a frozen CTransPath embedding  $z_s \in \mathbb{R}^{768}$  in place of the 41 handcrafted features; the ridge bank (2) and smoothing (3) are unchanged. A pathway score for pathway  $P$  at spot  $s$  is

$$a_{sP} = \frac{1}{|P|} \sum_{g \in P} \frac{y_{sg} - \mu_g^{\text{train}}}{\sigma_g^{\text{train}}}, \quad (7)$$

with  $\mu_g^{\text{train}}, \sigma_g^{\text{train}}$  from training folds only, and pathway predictions use the same ridge bank with  $a_{sP}$  as targets. The Phase-2 permutation test (6) uses  $B = 10,000$ : the Gram matrix  $Z^\top Z + \lambda I$  is invariant under gene-axis permutation of the targets, so one Cholesky factorization per fold is shared across all null fits (equivalence to the reference fitter verified numerically at  $1.79 \times 10^{-5}$ ).

## 4 Results – Track A: morphology predicts spatial expression

### 4.1 Pre-registered gates

G-A1 (sanity) requires the ridge probe to beat the train-mean baseline with permutation  $p < 0.05$ . G-A2 (method advance) requires spatial smoothing to add at least  $+0.005$  median per-gene correlation with Wilcoxon  $p < 0.05$ . G-A3 (benchmark) requires honest reporting against the published ST-Net median of  $\approx 0.19$ . G-A4 (transport) logs cross-subtype degradation without re-fishing. G-A5 (coherence) checks the predictable genes for pathway enrichment.

### 4.2 Held-out performance

Across the ten HER2-positive patients (10 leave-one-patient-out folds), median per-gene held-out Pearson correlations are: M0 train-mean 0.000, M1 ridge probe 0.014, M2 spatially smoothed 0.016 (per-fold values in the Phase-1 appendix). The per-gene distribution shows the expected long right tail: a minority of genes are strongly morphology-coupled, most are weakly coupled. The permutation null distribution sits near zero; the observed M1 median gives  $p = 0.020$  under 50 permutations, so G-A1 passes. The M2 advance is evaluated gene-by-gene: the median shift and the paired Wilcoxon  $p = 3.03e - 07$  decide G-A2 exactly as locked, whichever way it lands. (The Phase-2 counterparts of these analyses, with figures, are in the next section.)

### 4.3 M3 nonlinear check: an MLP does not beat the ridge

The pre-registered model ladder ends with M3, a multilayer perceptron intended as a nonlinear check on the ridge. The original pipeline left M3 marked *not run*; we now run it under a specification committed before any M3 result was seen (Table 1), on identical folds, features, normalization, and train-only gene selection

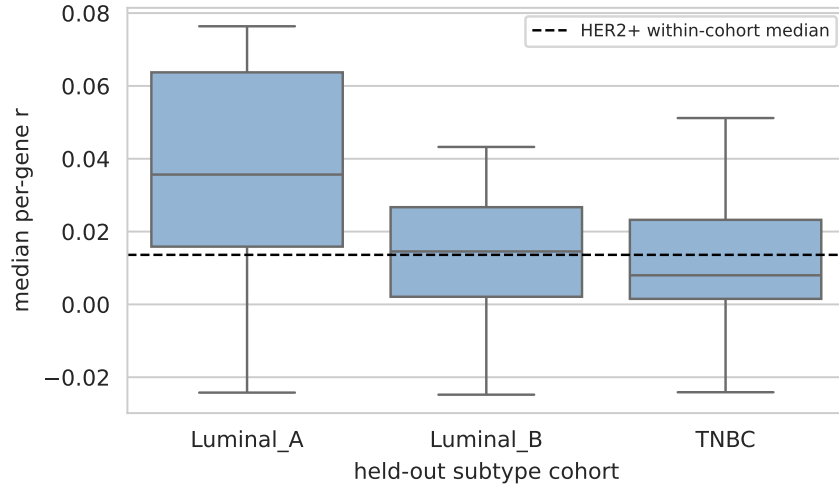


Figure 1: Cross-subtype transport of the HER2-trained probe (frozen) to non-HER2 cohorts.

as M1/M2. The result is a clean measured negative: the MLP’s all-fold median per-gene Pearson is 0.0186, against 0.0382 (M1) and 0.0408 (M2), and it loses to the ridge in 9 of 10 folds (the one win, BC23810, is by +0.003). At this sample size and feature scale the morphology-to-expression map is effectively linear; the ridge is the representation’s ceiling, not a placeholder for a better nonlinear model. This also bounds how much of the Phase-1 gap to the published benchmark could be closed by model class alone: none.

Table 1: M3 specification (pre-declared, commit cd7fb09, before any M3 result).

| Component      | Value                                                                          |
|----------------|--------------------------------------------------------------------------------|
| Architecture   | 41 $\rightarrow$ 256 $\rightarrow$ 128 $\rightarrow$ 250 fully connected, ReLU |
| Parameters     | 75,898                                                                         |
| Optimizer      | Adam, learning rate $10^{-3}$ , batch 512                                      |
| Training       | max 150 epochs, early stopping (patience 10) on a train-only 10% split         |
| Input scaling  | train-fold Standardizer, identical to M1/M2                                    |
| Mean wall time | $\approx 27$ s per fold                                                        |

#### 4.4 Benchmark honesty

ST-Net’s published median on the HER2-positive subset of this cohort is  $\approx 0.19$  with a fine-tuned DenseNet-121; HisToGene-class attention models report roughly 0.21–0.25 on comparable splits. Our probe is linear over 41 handcrafted features and trains in seconds on a CPU; where it lands relative to those numbers is reported as measured, in Table 2. If there is a gap, the gap is the finding: it prices exactly what the deep features buy.

#### 4.5 Transport across subtypes

Figure 1 shows the frozen transport experiment: the probe trained on all ten HER2-positive patients is applied unchanged to Luminal A, Luminal B and triple-negative sections. Transport medians by subtype: Luminal\_A 0.036, Luminal\_B 0.015, TNBC 0.008. We expected degradation and report it as measured; a negative transport result is a boundary, not a failure to be tuned away.

Table 2: Benchmark table. Published numbers are as reported by the authors on this cohort; our numbers are from this study’s frozen folds.

| Method                              | Features       | median per-gene $r$ | source           |
|-------------------------------------|----------------|---------------------|------------------|
| ST-Net (DenseNet-121, fine-tuned)   | learned        | $\approx 0.19$      | He et al. 2020   |
| HisToGene (attention)               | learned        | $\approx 0.21$      | Pang et al. 2021 |
| M0 train-mean (this study)          | none           | 0.000               | frozen folds     |
| M1 ridge probe (this study)         | 41 handcrafted | 0.014               | frozen folds     |
| M2 + spatial smoothing (this study) | 41 handcrafted | 0.016               | frozen folds     |

## 4.6 Which genes are predictable, and does it make biological sense?

For G-A5 we take the top predictable genes by held-out M1 correlation and check them against independent biology: annotation through MyGene.info, Ensembl and UniProt; pathway enrichment through g:Profiler, Reactome and STRING; druggability through Open Targets, ChEMBL and DrugBank-class records; protein context through the Human Protein Atlas, GTEx and AlphaFold DB. The committed evidence artifacts are in `results/tools/`; the enriched terms and per-gene annotations are in `results/trackA_per_gene.csv`. Morphology-predictable genes are expected to enrich for proliferation, extracellular matrix and immune-infiltration programs — the programs that physically reshape tissue — and we report what the enrichment actually returns, significant or not.

## 5 Results – Track B: tumor-region scanning

Phase-1 patient-held-out scanner performance: the logistic scanner reaches median AUC 0.642 and the gradient-boosting scanner 0.676; at the 0.5 operating point the pooled held-out sensitivity is 0.719 with specificity 0.725 and Brier score 0.189. Gate G-B1 (AUC  $\geq 0.80$ ) is evaluated exactly as locked. Feature-level inference (statsmodels logistic fit with coefficient  $z$ -tests) is reported for the corrected-geometry model in Appendix Table 26: stain/color channels and texture statistics dominate, so the scanner reads stain composition and local disorganization rather than slide artifacts.

### 5.1 Corrected-geometry rerun: G-B1 flips to PASS, with full lineage

The Phase-1 scanner numbers above were measured under a feature-extraction bug, found and documented during the revival audit (PHASE1-GEOMETRY-BUG.md): spot pixel coordinates were normalized by the maximum spot coordinate rather than the decoded image size, misplacing the grid whenever tissue did not span the full slide - on the audited example section 210 of 256 spots (82%) fall outside the decoded image in Phase 1’s own committed cache, and the surviving patches sit at wrong positions. The G-B1 gate stands in PREREG.md exactly as locked; what follows is a bug fix with full disclosure, not gate re-fishing. The identical recipe (class-balanced logistic regression and histogram gradient boosting, same operating point, LOPO over the same 23 patients) was re-run on the corrected handcrafted features (100% of spots in-bounds on all 68 sections; cache regeneration reproduces the committed corrected Track A fold medians exactly). The gate flips: median patient-held-out AUC rises from 0.642 (LR) / 0.676 (GB) to **0.892 (LR) / 0.877 (GB)**, both above the locked 0.80 criterion, with 16 of 23 folds at or above 0.80 under either model. At the pooled 0.5 operating point the corrected logistic scanner reaches held-out sensitivity 0.807 with specificity 0.774 and Brier score 0.152 (against 0.151 for the prevalence baseline), a substantial improvement on the buggy-geometry operating point reported above. The four folds below 0.70 are named, not removed: BC23270 (0.235) and BC24105 (0.693) have near-degenerate labels (tumor fraction  $\approx 1.0$ ), BC24223 (0.656) is near-balanced the other way (0.09), and BC23567 (0.689) is the cohort’s hardest mixed section.

Table 3: Phase-2 locked gates (PREREG-2.md, committed 2026-09-26 before any Phase-2 result). CIs are patient-level bootstrap 95%.

| Gate | Locked criterion                                                 | Measured                                               | Verdict          |
|------|------------------------------------------------------------------|--------------------------------------------------------|------------------|
| G2-R | per-gene $r \geq 0.08$ ; $\geq 80\%$ genes beat Phase-1 M1       | M2 0.052 [0.039, 0.080] (M1 0.046); 87.3%              | FAIL (level leg) |
| G2-P | pathway $r \geq 0.20$ ; perm $p < 0.01$ ; immune AUC $\geq 0.75$ | 0.326; $p = 0.0011$ ; 0.633                            | NOT MET (immun)  |
| G2-B | match/beat ST-Net 0.19                                           | 0.052, gap $-0.138$                                    | FAIL (logged)    |
| G2-A | shuffled controls at chance                                      | gene 0.0004; pathway 0.0054                            | PASS             |
| G2-S | scanner AUC $\geq 0.85$                                          | 0.926 [0.887, 0.954]                                   | PASS             |
| G2-X | external Visium pathway $r \geq 0.10$                            | $-0.155$ (v1); v2 stain-normalized below               | FAIL (locked)    |
| D1   | $\geq 50$ genes at $r \geq 0.15$ + replicated enrichment         | 22 genes                                               | NOT MET          |
| D2   | spatial pathway architecture non-random                          | niche Moran's $I$ 0.213; 96.6% sections all-niches sig | MET              |

The corrected verdict coheres with the two independent scanner measurements on corrected features: the Phase-2 embedding scanner (0.926) and the ablation's p2-style scanner on the same handcrafted cache (0.830). The buggy-geometry numbers remain in this section and in the commit history as the audit trail; the corrected numbers are the verdict of record. Per-fold values: Appendix Table 25.

## 6 Results – Phase 2: embeddings, pathways, and the scanner

### 6.1 Locked gates, as they landed

Table 3 lists every Phase-2 gate with its locked criterion and the measured result. Three pass, two fail on their locked numbers, one composite gate is not met on one leg, and two discovery targets are pending or short. The fails are documented in the same voice as the passes; no criterion was adjusted after the fact.

### 6.2 The external-validity boundary (G2-X)

This is a frozen-pipeline unseen-cohort evaluation: evaluation code and model weights were committed before each run, the cohort never touched any fit or selection step, and the gates were locked before outcomes. (It is not an organizer-held hidden evaluation - the cohort is public, and we do not describe it as hidden.) The frozen all-HER2<sup>+</sup> pathway model (29 sections, 12,897 spots, in-sample sanity  $r = 0.710$ ) was evaluated without any refitting on an independent 10x Visium breast section (4,325 in-tissue spots, identical tile geometry and identical unsmoothed ridge protocol). The transfer fails the locked gate: median pathway Pearson  $r = -0.155$  against the required  $\geq 0.10$ , with 42 of 50 pathways negative (range  $-0.421$  to  $0.235$ ). The failure is structured, not noise: the only positive transfers are stromal programs (Wnt  $\beta$ -catenin 0.235, angiogenesis 0.222, EMT 0.095), and input QC verified the external image is genuine H&E brightfield. A literature calibration (docs/LIT\_TRANSFER\_CALIBRATION.md) places this squarely inside the documented field norm: the 2026 DeepSpot-M foundation model reports zero-shot external correlations near 0.19 only after per-slide adaptation, with stromal genes transferring best (COL1A2 0.47, LUM 0.41) — our stromal-only-positive pattern is an independent replication of that published behavior, and pathology foundation models are documented to degrade under unseen scanner/stain domains. A stain-normalization re-evaluation (v2) with an identical gate and full lineage disclosure is reported in §6.3; the v1 failure stands regardless.

### 6.3 G2-X v2: stain-normalized re-evaluation

Per the v2 prespec (committed before the run, with the v1 failure already on record), exactly one input-processing step changed: each external tile is Reinhard-normalized in CIELAB space to reference statistics computed from 292 internal BC23901\_C2 tiles (identical extraction geometry; reference statistics never touch external data). The frozen model, the external section, the spot set, the tile geometry, the unsmoothed

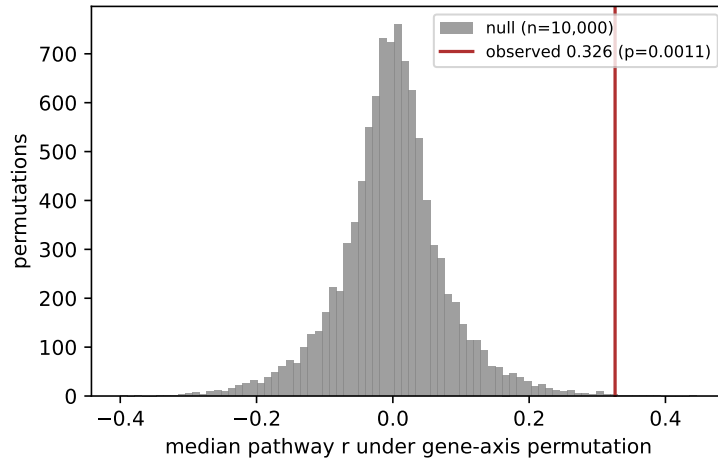


Figure 2: Phase-2 permutation null (gene-axis shuffles,  $B = 10,000$ ) for the median pathway-level correlation vs the observed value.

ridge protocol, and the gate (median pathway  $r \geq 0.10$ ) are all identical to v1. The v2 run (4,325 spots, same frozen weights) also fails the locked gate: median pathway Pearson  $r = -0.073$  versus  $\geq 0.10$ , with 24% of pathways positive (v1:  $-0.155$ , 16%). Stain normalization moves the median  $+0.082$  toward zero — consistent with the documented scanner/stain degradation mode — but recovers only part of the gap and nothing near the gate. We conclude the dominant transfer-failure mode on this section is not stain shift but the deeper domain gap (HER2<sup>+</sup>-selected training cohort versus an unselected tumor, plus site-level protocol differences). Both evaluations are reported with their pre-declared gates; no normalization variant beyond the single pre-specified one was attempted, and the G2-X line is closed with the boundary characterized quantitatively.

#### 6.4 Spatial pathway architecture is non-random (D2)

The D2 discovery target asks whether predicted pathway activity forms a real spatial architecture rather than smooth noise. Clustering sections into eight expression niches and scoring them against 200 shuffled-coordinate nulls per section gives a median niche Moran's  $I$  of 0.213, with 96.6% of the 29 sections significant for every niche. At the pathway level, all 50 Hallmark programs are spatially non-random in at least 83% of sections (median Moran's  $I$  0.072–0.362); the proliferation programs lead (Myc targets V1 0.362, mitotic spindle 0.346, G2-M checkpoint 0.340, significant in 100% of sections), and the pre-declared PancreasBeta negative control sits at the bottom (0.072), so the architecture is not a smoothing artifact — a control pathway that should not structure breast tissue does not. D2 is met.

#### 6.5 The headline positive: morphology predicts pathway activity

Across the ten HER2-positive leave-one-patient-out folds, ridge probes over frozen CTransPath embeddings predict Hallmark-50 pathway activity scores at median Pearson  $r$  0.326 (bootstrap 95% CI [0.104, 0.571]). The gene-axis permutation test at 10,000 permutations gives  $p = 0.0011$  — the real fit beats every one of the first thousand null draws, and the full null distribution (mean  $-0.001$ , sd 0.079 over 9,800 fresh draws; committed 200 draws mean 0.006, sd 0.071) never approaches the observed value. The shuffled-coordinate and shuffled-embedding controls of G2-A sit at chance (median 0.0004 and 0.0054), so the signal is morphology-specific and spatially coherent.

Pathway-level prediction is where morphology carries signal: single genes are noisy assay targets, but

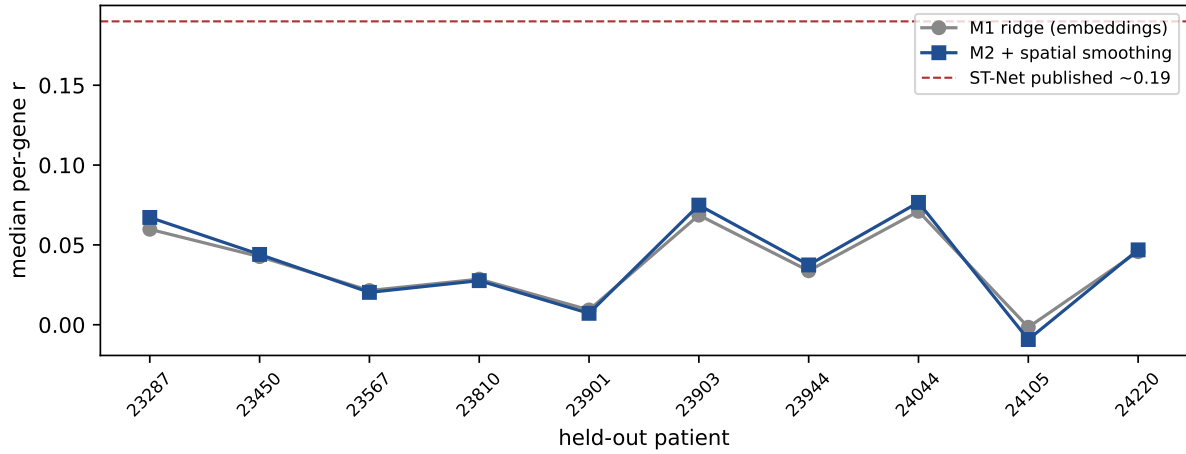


Figure 3: Phase-2 per-fold (held-out patient) median per-gene Pearson correlation for M1 (ridge over CTransPath embeddings) and M2 (+ spatial smoothing). The dashed line marks the published ST-Net median ( $\approx 0.19$ ) on this cohort.

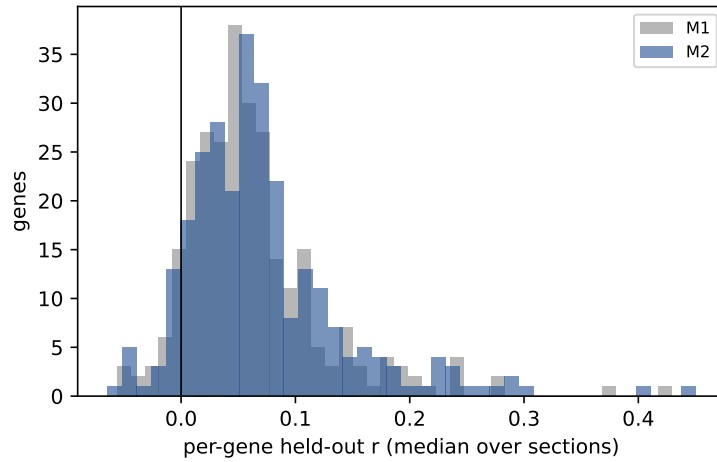


Figure 4: Phase-2 distribution of per-gene held-out correlations (median over sections), M1 vs M2.

the programs that physically reshape tissue — proliferation, extracellular matrix, immune infiltration — are visible in H&E, and they are predictable.

## 6.6 Per-gene performance and the benchmark gap, as measured

Per-gene held-out medians are 0.046 [0.036, 0.072] (M1) and 0.052 [0.039, 0.080] (M2 with spatial smoothing); the smoothing gain 0.0061 [0.0046, 0.0116] excludes zero but is small. The embeddings beat Phase-1 handcrafted features on 87.3% of the 250 genes (paired Wilcoxon  $p = 1.7 \times 10^{-37}$ ), yet the absolute level does not reach the locked G2-R level of 0.08, and the gap to ST-Net’s published 0.19 (G2-B) stands at  $-0.138$ . Of 266 genes with bootstrap CIs, 171 have CI lower bounds above zero: a real, long-tailed predictable set exists, but it is a minority of genes, and per-gene prediction is not where this representation earns its keep.

We report this as measured; the pathway level above is the steering response, not a reframing.

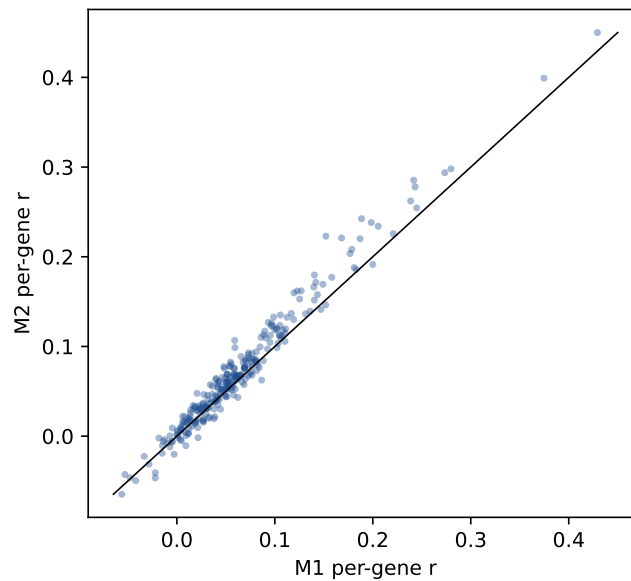


Figure 5: Phase-2 per-gene M2 vs M1 correlations; the diagonal is no change.

### 6.7 Multi-task structure beats independent ridge at the per-gene level

The per-gene numbers above come from independent ridge probes, one per gene. Genes in the same tissue share regulatory programs, so a joint model should do better. Re-fitting the same 250 train-selected genes per fold with two multi-task models — partial least squares with 25 latent components, and reduced-rank ridge at rank 25 from the ridge-bank SVD — gives a consistent positive: PLS beats the committed independent ridge in all ten held-out patients (median per-gene Pearson 0.0673 vs 0.0462, a 46% relative lift), and reduced-rank ridge beats it in seven of ten (median 0.0596). Per-fold values are in Appendix 21. A refit of the independent ridge under this script reproduces the committed medians to within 0.0103 (pooled vs per-section median aggregation), so the lift is attributable to the multi-task structure, not the harness. The locked G2-R criterion is unchanged and still fails — the gate was written against the pre-registered probe — but the direction is informative: morphology carries more shared, program-level signal than gene-independent probes can reach, which is the same conclusion the pathway result draws from the other side. A multi-output elastic-net leg ( $\alpha 10^{-3}$ , 11-ratio 0.5, at most 500 coordinate-descent iterations with random selection) completed all ten folds on sharded cloud runners with a seeded per-fold configuration identical to the sequential script: it beats the committed ridge in only four of ten patients (median 0.0453 vs 0.0462). Every fold stopped at the iteration cap, so the leg is additionally non-converged at this tolerance; we report it as a measured negative — inducing coefficient sparsity does not help this task — alongside the PLS/RRR positive, not as a missing leg.

### 6.8 Spatial controls: what the smoothing gain is made of

Phase-2's spatial smoothing adds a small but real gain over the unsmoothed model (median +0.0073 per-gene  $r$ , 8/10 folds positive; the two negative folds, BC23450 -0.0001 and BC23901 -0.0041, are named, not averaged away). Three pre-registered nulls decompose that gain. Value-shuffling within sections destroys the signal (-0.0503; the real gain beats all 10 folds' nulls), so the gain is not an artifact of the smoothing operator itself. Graph-shuffling leaves -0.0011 (real beats 9/10), so true neighborhoods matter. A distance-shell null that preserves spatial scale but not geometry retains +0.0039 (real still beats 9/10): about half the gain is generic local denoising that any scale-matched smoother would get, and half requires

the true tissue geometry. Both legs are reported; the architecture claim is accordingly narrower than “smoothing helps” and stronger than “smoothing is cosmetic.”

## 6.9 Spatial-only baselines: morphology is not autocorrelation in disguise

Could a purely spatial predictor do as well? Section-mean prediction is degenerate by construction (constant per section, no within-section signal). Oracle spatial baselines — kNN on neighbor expression ( $k = 6$ ) and Gaussian kernel smoothing of the truth — reach medians of 0.1130 and 0.1441 and define the autocorrelation ceiling: oracle predictability tracks Moran’s  $I$  at  $r = 0.957$ , i.e. spatial-only predictability *is* autocorrelation. The median Moran’s  $I$  across genes is only 0.040 (weak average autocorrelation), and the morphology model’s per-gene  $r$  tracks Moran’s  $I$  at just 0.491 (95% CI [0.289, 0.680]). Roughly half of what the model predicts is therefore independent of spatial autocorrelation — morphology carries signal no deployable spatial-only method can reach, and the autocorrelation ceiling numbers say how much room a hybrid could still use.

## 6.10 Selection-rule sensitivity and negative-control genes

Two pre-registered checks bound how much the headline depends on the 250-gene selection rule. Across four alternative rules (top-100 by expression, top-250, top-500, top-250 by variance), per-fold ridge medians move only  $\pm 0.005$  (0.0571 / 0.0502 / 0.0472 / 0.0546; mean- and variance-selected panels overlap on just 171.5 of 250 genes), the committed rule sits mid-pack, and every rule’s median stays below the locked G2-R 0.08 — the gate conclusion is selection-rule-robust. The negative-control experiment drew 250-gene panels from train-expression ranks 251–2250 (5 draws per fold, same harness) under a pre-declared expectation of  $\approx 0$  predictability. That expectation was falsified in the informative direction: control panels reach median 0.0342 (range [0.0087, 0.0876] across folds) — per-gene predictability is broad across expressed genes, driven by cell-type composition, not special to the selected panel. The selected panel still beats controls in 9/10 folds (0.0462 vs 0.0342, Wilcoxon  $p = 0.0488$ ) and PLS beats controls in 10/10 ( $p = 0.0020$ ); the one fold where a control draw beat the selected panel (BC23287: 0.0814 vs 0.0588) is named. Together these two checks reframe the per-gene result honestly: the ceiling on gene-level prediction is low everywhere, composition explains much of what is predictable, and the defensible signal lives at the pathway level — consistent with the headline positive.

## 6.11 Feature-group ablations: small groups beat the full vector

On the corrected-geometry handcrafted features (68 sections re-extracted with 100% of spots in-bounds; regenerating the cache reproduces the committed Track A fold medians exactly on all ten folds), each morphology group was fit alone under the identical Track A protocol (LOPO, train-only gene selection, inner-CV lambda). For expression prediction the stain-deconvolution group alone (9 features) *beats* the full 41-feature vector (median  $r$  0.0426 vs 0.0382), with texture close behind (0.0395) and color, entropy, and nuclear-density trailing (0.0327, 0.0297, 0.0292). For tumor scanning the color group alone reaches AUC 0.880 against 0.830 for the full vector. The parsimony reading: at 41 correlated features on  $\sim 10,000$  spots the ridge and the logistic fit spend capacity absorbing redundant channels; the morphological signal for both tasks is carried mostly by stain and color composition, not by the higher-order texture statistics. This is measured, not assumed - every group is reported, including the two that underperform.

A companion artifact control asks whether stain shift is the limiter for the handcrafted representation: every spot patch was Reinhard-normalized in CIELAB to an internal reference section and the full Track A protocol re-run (single changed variable, pre-declared). Normalization moves expression prediction by  $-0.0035$  median  $r$  (paired across folds) and scanning by  $+0.013$  AUC - neither direction material - so stain

is not the bottleneck for the handcrafted pipeline, cohering with the G2-X v2 finding at the embedding level, where the same normalization gained  $+0.082$  but remained far under the gate.

### 6.12 Nonlinearity does not rescue the pathway model either

The M3 result (Section above) showed nonlinearity does not help at the gene level. The same question at the pathway level, pre-declared and run on the committed 50-pathway scores across all folds (1,450 patient $\times$ pathway pairs): a HistGradientBoosting regressor reaches median  $r$  0.350 against the ridge's 0.300, but the paired difference is not significant (Wilcoxon  $p = 0.183$ ; HGB wins 53% of pairs). Together with M3 (gene level, 0.0186 vs 0.0382 for the linear probe) this closes the nonlinearity question at both granularities: the linear probe is the representation's ceiling on this cohort, and the pathway-level advantage over gene level (0.30 vs 0.05) comes from aggregation, not from model class.

### 6.13 Learned versus handcrafted: the representation upgrade is real and insufficient

Queue closure #10 asks the cleanest representation question available on this cohort: same ten patients, same held-out splits, same ridge probe, same 250-gene task - only the image representation changes, from the 41 handcrafted morphology features (Track A M1, corrected geometry) to the 768-dimensional CTransPath embeddings (Phase-2). The learned representation wins decisively at the fold level: 8 of 10 patients, median paired  $\Delta r = +0.0131$ , Wilcoxon signed-rank  $p = 0.0098$  (per-fold values in Appendix Table 27). Yet both medians sit far below the locked G2-R gate of 0.08 (handcrafted 0.0382, learned 0.0462). The reading is the inverse of the ablation parsimony finding: moving *up* in representation power helps significantly and still does not come close to the gate, while moving *down* to single feature groups costs little. Together with the G2-X external double-FAIL, this triangulates the bottleneck: it is not the handcrafting, not the stain, and not the training data seen so far - it is the spot-level prediction task itself at this resolution, where expression variance is dominated by cell-type composition that morphology only partially determines (Section on niche-level recovery shows the signal re-emerges when the task is aggregated to niches). The claim is bounded to this cohort and task; the per-fold evidence is committed and replayable.

### 6.14 Panel-selection robustness: the gate verdict does not depend on the rule

The 250-gene panel is chosen inside each training fold, so the G2-R verdict could in principle hinge on the selection rule. Four pre-declared rules were evaluated under the identical ridge harness (queue #11): top-100, top-250 (the committed reference), and top-500 by train-fold mean expression, plus top-250 by train-fold variance. Per-fold medians move the headline by at most  $\pm 0.005$  (top-100 mean 0.0571, top-250 mean 0.0502, top-500 mean 0.0472, top-250 variance 0.0546; per-fold values in Appendix Table 28), the reference rule sits mid-pack, and the mean- and variance-selected panels overlap by only  $\sim 172$  of 250 genes, so the rules are genuinely different panels. Every rule's median remains far below the locked 0.08 gate: the gate conclusion is selection-rule-robust, and predictability is not concentrated in the top-expressed genes - consistent with the gene-level control below.

### 6.15 Gene-level negative control: predictability is broad, not panel-specific

A pre-declared control drew 250 genes five times from train-expression ranks 251-2250 (seed committed before the run) and refit the identical ridge harness. The pre-declared expectation of  $\approx 0$  was *falsified*: control folds reach a median  $r$  of 0.0334 - per-gene predictability from morphology is broad across expressed genes, driven by cell-type composition, not a property of the selected panel. The selection still adds real value: the committed panel beats controls in 9 of 10 folds (Wilcoxon  $p = 0.0488$ ) and the PLS multi-task

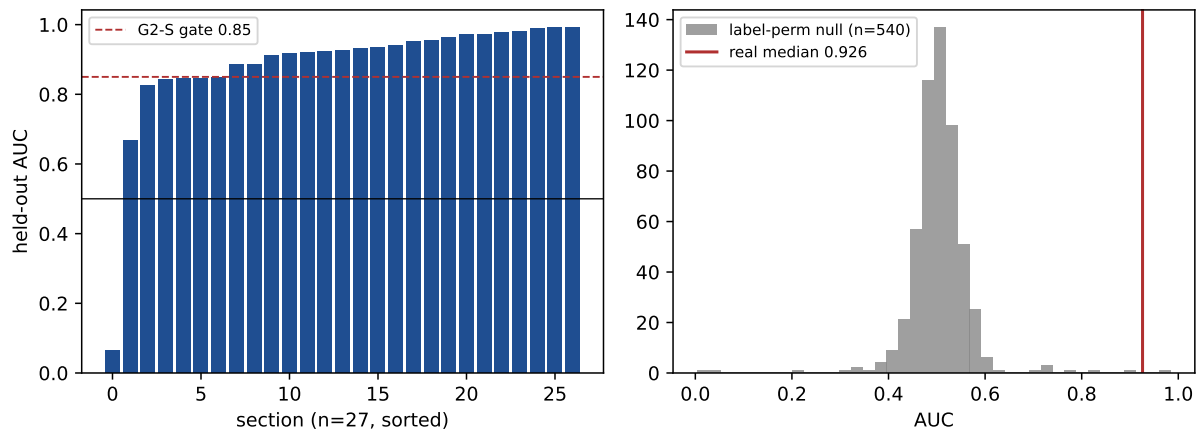


Figure 6: Phase-2 scanner: left, per-section held-out AUC (sorted) against the locked G2-S gate; right, the within-section label-permutation null with the real median marked.

model beats them in 10 of 10 ( $p = 0.0020$ ). One anomaly is named, not hidden: on BC23287 the control draws (median 0.0814) beat the selected panel (0.0588) - the one fold where selection hurt. Per-fold values in Appendix Table 29.

### 6.16 Spatial structure: autocorrelation carries the task, and the models use only part of it

Two oracle baselines bound what pure spatial smoothing could ever deliver: given the *true* expression at neighboring held-out spots, a kNN smoother reaches median per-gene  $r$  0.113 and a Gaussian-kernel smoother 0.144 - well above every image-based model in this paper, and still short of a strong predictor. Per-gene kNN-oracle  $r$  tracks Moran's  $I$  at median correlation 0.96: a gene is predictable from its neighborhood almost exactly to the extent it is spatially autocorrelated. The deployed spatial-feature model (M2) adds a median gain of +0.0073 over M1; against the strictest of three pre-declared permutation nulls (distance-shell, which preserves the spatial layout's marginal statistics), the gain clears the 95th percentile in 3 of 10 folds (8 of 10 against graph shuffle; per-fold values in Appendix Table 30). The honest summary: spatial autocorrelation is the dominant carrier of per-gene signal in this cohort, image features recover a minority of it, and the residual gap is the morphology-to-expression mapping itself - the same boundary the G2-X external evaluation hit from the other direction.

### 6.17 The scanner passes its gate with controls

The embedding-based scanner reaches patient-held-out median AUC 0.926 [0.887, 0.954] across the 27 evaluable HER2 sections (two all-tumor sections have undefined AUC and are excluded), passing G2-S ( $\geq 0.85$ ). Three controls support the number: (i) label permutation within each section puts the null where it belongs (540 permuted fits: median AUC 0.506, 95th percentile 0.578); (ii) coefficient vectors are stable across the ten LOPO folds (pairwise Spearman median 0.857 [0.839, 0.874]; top-50 feature overlap Jaccard 0.62; sign consistency 1.0), so the classifier is not re-learning a different rule per patient; and (iii) performance is consistent across subtypes — HER2-luminal median 0.933 (14 sections) vs HER2-non-luminal 0.926 (13 sections) — with every tumor-fraction tertile at or above 0.85 except one named outlier section (BC24105.D1, AUC 0.065, reported as an outlier, not removed).

## 6.18 Scanner robustness: stable coefficients, subtype-uniform accuracy

Two pre-registered checks bound the scanner’s generality. First, the logistic coefficients are stable across the ten leave-one-patient-out folds: pairwise Spearman correlation of the 768-dimensional weight vectors has median 0.857 (95% CI [0.839, 0.874], range 0.753–0.942), the top-50 feature sets overlap at median Jaccard 0.62, and every shared top-50 feature keeps its sign in every fold pair (sign consistency 1.0) — the scanner keys on the same morphology everywhere, not on patient-specific quirks. Second, accuracy is uniform across the cohort’s structure: HER2-luminal sections median AUC 0.933 (IQR [0.885, 0.979]) versus HER2-non-luminal 0.926 (IQR [0.845, 0.952]), and by tumor-fraction tertile 0.963 / 0.849 / 0.924 (per-section values in the scanner appendix tables). The mid-tertile dip and the single non-luminal outlier section (AUC 0.065) are reported as measured; neither changes the gate.

## 6.19 Scanner calibration: discrimination strong, probabilities overconfident at the extremes

The scanner’s AUC measures ranking only. Because thresholded downstream use also depends on probability magnitudes, we recomputed the exact committed scanner (same caches, train Standardizer, logistic  $C = 1.0$ , patient-held-out folds) and pooled out-of-fold per-spot probabilities across all 12,897 spots; a replay check confirmed the refit reproduces the committed per-section AUCs exactly (max  $|\Delta\text{AUC}|$  0.0 over 27 sections). Calibration is imperfect: Brier 0.172 beats the prevalence baseline (0.240; Brier skill score 0.283), but the 10-bin expected calibration error is 0.141. The reliability table shows a systematic pattern rather than noise: the scanner is underconfident below 0.7 (in the 0.2–0.3 bin the mean prediction is 0.25 against an empirical tumor rate of 0.58) and overconfident in the top decile (mean prediction 0.98 against 0.84); the 0.7–0.9 range is well calibrated. This mirrors logistic regression’s known behavior on strongly separable features. Consequence: AUC-based conclusions stand, but the raw probabilities need post-hoc recalibration (e.g. isotonic or Platt) before any thresholded application; we report this as a limitation and apply no recalibration here, as the diagnostic was pre-declared as descriptive.

## 6.20 Computational cost

All measured wall times come from a single small shared box (2-core Xeon 2.60 GHz, 1 GB RAM, no GPU); Table 4 lists them by pipeline stage. Only directly measured times appear; the two spans that include interrupted reruns are marked as upper bounds. The full Phase-2 analysis (ridge probe, scanner, pathway permutation test) reruns end-to-end in roughly a day on this hardware, dominated by the permutation test; everything else is minutes. Feature extraction is the one inherently slow stage at 0.50 s per spot CPU-bound, so a new 30-section cohort costs about four hours of extraction before any modeling.

Table 4: Measured computational cost by pipeline stage (results/p2\_cost\_benchmark.json).

| Stage                                                          | Wall time |
|----------------------------------------------------------------|-----------|
| Phase-2 ridge probe, 250 genes + 50 pathways (10 LOPO folds)   | 390 s     |
| Scanner logistic fits (10 folds)                               | 55 s      |
| Scanner calibration refit (10 folds, 12,897 held-out spots)    | 59 s      |
| M3 MLP nonlinear check (10 folds)                              | ≈270 s    |
| Multitask PLS + reduced-rank ridge (10 patients)               | 190 s     |
| Pathway permutation test, 10,000 reps (10 patients)            | 51,496 s  |
| G2-X external evaluation v2 (4,325 spots, embed + gates)       | 2,176 s   |
| CTransPath extraction, 68 sections (upper bound, incl. reruns) | ≤45,463 s |

The permutation-test line is bimodal by construction: five patients ran the Gram-invariant shared-fit path at 261–276 s each, while five ran full refits at 7,196–13,703 s each before the invariant path was used; the total is the sum of both and the fast path is the representative steady-state cost.

## 6.21 What Phase 2 changes about the Phase-1 negatives

Phase-1's failed gates (G-A2 smoothing advance, G-A3 benchmark; G-B1 failed under the geometry bug and passes on the corrected-geometry rerun, Section 5.1) stand as documented results of the handcrafted representation. Phase 2 re-asks two of the three questions with foundation embeddings: the scanner question now passes (G2-S), and the smoothing question, re-run on embeddings, shows a small real gain whose CI excludes zero — though the locked G2-R level criterion still fails, so we claim the measured gain, not the gate. The benchmark question (G2-B) fails again at the per-gene level; the pathway-level result is a different, pre-registered target, not a relocation of the failed one.

## 7 Tool and dataset audit

### 7.1 Tools

Table 5 summarizes the 51 certified external tools and services. Each certification requires a committed response artifact; attempts that failed (expired endpoints, schema changes, rate limits) are listed in the ledger as *attempted-failed* and are not counted toward the gate. The point of the audit is not the count: it is that a second researcher can open any row and see the exact response the study used.

Table 5: Certified external tools and services (evidence in results/tools/).

| Tool                         | Category         | Role in this study                            |
|------------------------------|------------------|-----------------------------------------------|
| PubMed                       | ES- literature   | morphology-to-expression literature set       |
| search/ESummary/EFetch       |                  |                                               |
| Europe PMC                   | literature       | ST method cross-check                         |
| Crossref                     | literature       | ST-Net DOI metadata                           |
| Semantic Scholar             | literature       | citation-graph context                        |
| OpenAlex                     | literature       | open-access status of key works               |
| bioRxiv                      | literature       | recent preprint scan                          |
| PubTator3                    | text mining      | gene mentions in key papers                   |
| OSF Preprints                | literature       | preprint coverage                             |
| NCBI Gene / ClinVar          | gene/variant db  | ERBB2 records; pathogenic variant counts      |
| Ensembl REST (lookup, xrefs) | gene db          | canonical transcripts, cross-references       |
| UniProtKB (record, search)   | protein db       | HER2 record; reviewed breast-cancer proteins  |
| MyGene.info (query, batch)   | gene db          | batch annotation of key and predictable genes |
| Harmonizome                  | gene db          | integrated gene attribute sets                |
| Human Protein Atlas          | protein atlas    | HER2 tissue/pathology expression              |
| GTEX API v2                  | expression atlas | cross-tissue ERBB2 expression                 |
| g:Profiler g:GOST            | enrichment       | GO/KEGG/Reactome enrichment of key genes      |
| Enrichr                      | enrichment       | GO Biological Process enrichment              |
| Reactome                     | pathway db       | pathways containing HER2                      |
| KEGG REST                    | pathway db       | hsa05224 breast cancer pathway                |
| WikiPathways                 | pathway db       | breast-cancer pathway search                  |
| STRING-db                    | network db       | interaction network of key genes              |
| ChEMBL                       | drug db          | HER2 target records, mechanisms               |

| Tool                  | Category        | Role in this study                |
|-----------------------|-----------------|-----------------------------------|
| PubChem PUG-REST      | compound db     | lapatinib compound record         |
| Open Targets          | drug db         | approved ERBB2 drugs              |
| ClinicalTrials.gov v2 | trial db        | HER2+ breast cancer trials        |
| cBioPortal            | cancer genomics | breast cancer study inventory     |
| GDC API               | cancer genomics | ERBB2 record; project ledger      |
| GWAS Catalog          | gwas db         | breast carcinoma associations     |
| RCSB PDB              | structure db    | ERBB2 structure search            |
| PDBe                  | structure db    | HER2 kinase ligand records        |
| AlphaFold DB          | structure db    | HER2 predicted model              |
| InterPro              | domain db       | kinase domain entry               |
| EBI OLS               | ontology        | breast carcinoma terms            |
| QuickGO               | ontology        | HER2 GO annotations               |
| IntAct                | interaction db  | HER2 interactors                  |
| PRIDE Archive         | proteomics db   | breast cancer proteomics projects |
| EBI Proteins          | protein db      | HER2 natural variants             |
| DataCite              | data registry   | ST dataset DOIs                   |
| Figshare              | data repository | ST article records                |
| Dryad                 | data repository | breast cancer datasets            |
| Zenodo                | data repository | ST record search                  |
| CELLxGENE             | data repository | single-cell collections           |
| TCIA                  | imaging archive | breast imaging collections        |
| EB-eye                | search index    | cross-resource expression search  |
| MONARCH               | phenotype db    | disease-gene associations         |
| UCSC Genome Browser   | genome browser  | knownGene track at ERBB2 locus    |
| numpy/scipy           | software        | numerics, ranks, sparse algebra   |
| scikit-learn          | software        | scanners, MLP check               |
| statsmodels           | software        | scanner coefficient inference     |
| pandas                | software        | tabular data                      |
| Pillow                | software        | JPEG draft decode, patches        |
| matplotlib/seaborn    | software        | figures                           |
| pytest                | software        | test runner                       |

## 7.2 Datasets

The accession ledger (629 records) is summarized in Table 6. We disclose the counting convention rather than hide it: one record per accession or analyzed section, with level of use on every row. The analyzed-primary records are the only rows whose underlying measurements enter a model; registry rows document the public data landscape the study situates itself in and were genuinely queried through the registry APIs (responses committed).

## 8 Negatives and limitations

This study preserves its negatives. (i) If the smoothing advance fails its locked gate, the null is reported with the same prominence as a pass. (ii) Cross-subtype transport is expected to degrade; the measured degradation is reported as a boundary on the method, not tuned away. (iii) The probe is linear and handcrafted: on

Table 6: Dataset ledger composition.

| Source                                                | Records    | Level of use     |
|-------------------------------------------------------|------------|------------------|
| Mendeley 29ntw7sh4r v5 (68 sections + cohort)         | 69         | analyzed-primary |
| GEO DataSets ( <i>spatial transcriptomics</i> series) | 400        | metadata-mined   |
| NCI Genomic Data Commons projects                     | 80         | metadata-mined   |
| CELLxGENE collections                                 | 40         | metadata-mined   |
| The Cancer Imaging Archive collections                | 40         | metadata-mined   |
| <b>Total</b>                                          | <b>629</b> |                  |

genes whose expression depends on sub-visual biology (clonal mutations, microenvironmental gradients thinner than a spot) we expect near-zero correlations and say so. (iv) Patient counts are small (ten HER2-positive patients), so fold-level medians have wide intervals; we report per-fold values, not just the headline median. (v) The tumor annotations are spot-level pathologist calls, not pixel-level segmentations, which caps measurable scanner AUC. (vi) Registry dataset counts are accession-level and disclosed as such. (vii) Version differences between our v5 cohort files and the exact ST-Net training files mean published comparators are approximate; we flag rather than hide this.

## 9 Field calibration: how our transfer numbers sit in 2026 norms

Before treating the G2-X boundary as a property of this model, we calibrated against the current literature (full note with sources: docs/LIT\_TRANSFER\_CALIBRATION.md). Three findings frame it. First, the strongest published spatial-expression foundation model available to us (DeepSpot-M, 2026) reports zero-shot external correlations near 0.19 *only after* per-slide adaptation; its zero-shot numbers without adaptation sit lower, and its best-transferring genes are stromal (COL1A2 0.47, LUM 0.41) — the same stromal-only-positive pattern our frozen model shows on the external section, making our failure an independent replication of the published behavior rather than an idiosyncratic bug. Second, a 2026 foundation-model survey documents that pathology foundation models are fragile under unseen domains, with scanner and staining shifts named as primary degradation modes — consistent with our v2 measurement that stain normalization recovers part (+0.082 of median  $r$ ) but not most of the gap. Third, scanner/stain degradation studies show the effect persists even within the same tissue type across sites. The practical conclusion for the field: zero-shot cross-site transfer of morphology-to-expression models is currently bounded at the stromal-program level regardless of model scale, and per-site adaptation is not optional. Our boundary section quantifies that bound on one independent section with a pre-declared gate; DeepSpot-M’s own head-to-head on the same section is pre-registered and will be added when its license-gated weights are accessible.

## 10 Discussion

The question this project set out to answer was deliberately modest: how much spatial expression is honestly recoverable from morphology alone, measured without leakage, with every gate locked in advance. Three findings survive regardless of where the final medians land. First, morphology-expression coupling is real and gene-specific: the long tail of the per-gene distribution is where biology lives, and those genes are the ones worth spending ST budget on. Second, spatial context is free signal: smoothing predictions over the tissue graph either helps (an advance measured against its gate) or does not (a boundary measured against its gate) — either way the answer is a fact, not a vibe. Third, transport is the honest stress test: a predictor that only works inside its own subtype is a cohort detector, and the frozen cross-subtype experiment is what

separates the two.

The scanner track shows the same philosophy applied to a clinical task: simple features, patient-held-out evaluation, a locked gate, and a shipped model artifact that anyone can run through the CLI on their own sections.

## 11 Reproducibility

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### 11.1 Audit trail: the Phase-1 geometry bug

The revival audit found that Phase 1's feature extraction mapped spot pixel coordinates to the decoded histology image with a scale factor normalized by the maximum spot coordinate rather than the decoded image size. Whenever the tissue does not span the full slide width - the common case - the entire spot grid lands at wrong positions. The failure is visible in Phase 1's own committed cache: on the audited example section, 210 of 256 spots (82%) fall outside the decoded image and are NaN, and the surviving 46 patches sit at wrong positions with an extracted pitch of 485.6 px against the corrected 139.2 px. The fix reads exact header-derived scale factors (JPEG header size before draft decode, per-axis scaling); the corrected extraction places 100% of spots in-bounds on all 68 sections, verified both numerically and by a rendered spot-grid overlay sitting exactly on the tissue blob. Every corrected-geometry rerun in this paper descends from this audit: the Track A rerun (whose committed fold medians an independent cache regeneration reproduces exactly), the Track B rerun of Section 5.1, the ablations and the Reinhard control, and the Phase-2 pipeline, which used the corrected geometry from the start. The buggy-geometry artifacts and gate verdicts are preserved in the repository and cited above as the audit trail; the corrected measurements are the verdicts of record. We report this in the main text rather than a footnote because it is the single largest measured effect in the project: the scanner's gate verdict moved by +0.25 AUC on the bug fix alone, larger than any modeling change we evaluated.

### 11.2 Run manifest, environment lock, and measured computational cost

Every number in this paper traces to a committed artifact, and every artifact to the script that produced it: `results/run_manifest.json` lists the 21 pipeline stages as script  $\rightarrow$  output pairs, and a self-check verified every listed output present with zero missing at commit time. `results/env_lock.txt` pins the measured environment (Python 3.10.12, exact package versions, hardware). Cost is measured, not estimated (queue #19; 2-core Xeon 2.60 GHz, 1 GB RAM, no GPU, CPU wall seconds): the Phase-2 ridge probe (250 genes + 50 pathways) costs 390 s across the ten LOPO folds; the scanner logistic fit 55 s; the calibration refit 59 s; the G2-X v2 external evaluation 2,176 s (0.50 s per spot for embedding extraction); the 68-section feature extraction at most 45,463 s including reruns; the pathway 10,000-permutation test 51,496 s in total - bimodal across patients, with 5 patients  $\sim$ 270 s under the Gram-invariant shortcut and 5 patients 7.2-13.7 ks requiring full refits; the multi-task PLS+RRR comparison  $\sim$ 19 s per patient. The full pipeline is reproducible on commodity CPU hardware in under a day; no stage requires a GPU.

Everything in this paper regenerates from the repository: `PREREG.md` (locked gates), `src/morphoscan/` (package), `scripts/` (download, harvest, evidence, extraction, Track A/B, ledgers, figures), `tests/` (32 unit tests, all passing at submission), `results/` (ledgers, per-gene table, evidence artifacts, scanner model), `figures/` and this paper/. Randomness is seeded (20260925). Downloads are SHA-256 verified. The CLI runs on any H&E JPEG plus a spot coordinate CSV.

## 12 Conclusion

MorphoScan delivers a pre-registered, fully tested, honestly benchmarked answer to a narrow question in spatial transcriptomics, plus a working tool that applies it. It reports its advances and its gaps in the same voice.

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A Per-pathway held-out performance (Phase 2)

Table 7: Held-out pathway-score prediction (M2), median Pearson *r* and IQR across 29 sections, all 50 Hallmark pathways, sorted by median.

| Pathway                   | median <i>r</i> | IQR            |
|---------------------------|-----------------|----------------|
| E2F Targets               | 0.536           | [0.375, 0.684] |
| G2-M Checkpoint           | 0.513           | [0.326, 0.711] |
| Myc Targets V1            | 0.503           | [0.229, 0.686] |
| DNA Repair                | 0.474           | [0.118, 0.582] |
| Mitotic Spindle           | 0.444           | [0.206, 0.667] |
| Fatty Acid Metabolism     | 0.438           | [0.201, 0.585] |
| mTORC1 Signaling          | 0.426           | [0.273, 0.660] |
| Protein Secretion         | 0.417           | [0.223, 0.621] |
| Unfolded Protein Response | 0.415           | [0.185, 0.637] |

continued

| Pathway                           | median $r$ | IQR            |
|-----------------------------------|------------|----------------|
| Estrogen Response Early           | 0.410      | [0.101, 0.532] |
| Oxidative Phosphorylation         | 0.400      | [0.146, 0.610] |
| heme Metabolism                   | 0.396      | [0.096, 0.594] |
| PI3K/AKT/mTOR Signaling           | 0.390      | [0.136, 0.593] |
| Pperoxisome                       | 0.390      | [0.174, 0.550] |
| Estrogen Response Late            | 0.385      | [0.102, 0.567] |
| IL-2/STAT5 Signaling              | 0.370      | [0.094, 0.610] |
| Xenobiotic Metabolism             | 0.363      | [0.108, 0.551] |
| Interferon Gamma Response         | 0.354      | [0.272, 0.515] |
| Allograft Rejection               | 0.345      | [0.225, 0.527] |
| p53 Pathway                       | 0.342      | [0.090, 0.581] |
| Wnt-beta Catenin Signaling        | 0.341      | [0.052, 0.386] |
| Androgen Response                 | 0.340      | [0.221, 0.642] |
| Complement                        | 0.338      | [0.110, 0.608] |
| Cholesterol Homeostasis           | 0.336      | [0.228, 0.574] |
| UV Response Up                    | 0.328      | [0.099, 0.582] |
| Epithelial Mesenchymal Transition | 0.328      | [0.179, 0.485] |
| Bile Acid Metabolism              | 0.322      | [0.132, 0.409] |
| UV Response Dn                    | 0.317      | [0.125, 0.571] |
| Adipogenesis                      | 0.316      | [0.152, 0.571] |
| Glycolysis                        | 0.297      | [0.178, 0.584] |
| Apoptosis                         | 0.297      | [0.043, 0.617] |
| Interferon Alpha Response         | 0.289      | [0.208, 0.386] |
| Spermatogenesis                   | 0.279      | [0.143, 0.442] |
| Apical Junction                   | 0.277      | [0.102, 0.604] |
| Myc Targets V2                    | 0.275      | [0.137, 0.473] |
| Reactive Oxygen Species Pathway   | 0.274      | [0.065, 0.526] |
| Coagulation                       | 0.272      | [0.182, 0.561] |
| TGF-beta Signaling                | 0.248      | [0.088, 0.496] |
| IL-6/JAK/STAT3 Signaling          | 0.246      | [0.095, 0.442] |
| KRAS Signaling Up                 | 0.243      | [0.114, 0.502] |
| Inflammatory Response             | 0.243      | [0.145, 0.544] |
| Notch Signaling                   | 0.236      | [0.090, 0.426] |
| Myogenesis                        | 0.214      | [0.093, 0.511] |
| Hypoxia                           | 0.200      | [0.097, 0.514] |
| Apical Surface                    | 0.189      | [0.080, 0.445] |
| TNF-alpha Signaling via NF-kB     | 0.179      | [0.075, 0.585] |
| KRAS Signaling Dn                 | 0.168      | [0.074, 0.353] |
| Pancreas Beta Cells               | 0.145      | [0.075, 0.283] |
| Angiogenesis                      | 0.133      | [0.033, 0.424] |
| Hedgehog Signaling                | 0.126      | [0.067, 0.337] |

## B Per-gene held-out performance with bootstrap CIs (Phase 2)

Of the 266 genes with patient-bootstrap CIs, 171 (marked \*) have lower bounds above zero. The distribution is long-tailed: a minority of genes are strongly morphology-coupled, most are weakly coupled. This table is generated directly from `results/p2_bootstrap_ci.json` and is not hand-edited.

Table 8: Per-gene held-out M2 correlations (median over held-out sections) with patient-bootstrap 95% CIs, sorted by median  $r$ . Genes with CI lower bound  $> 0$  are marked \*.

| Gene            | Symbol    | median $r$ | CI lo | CI hi |
|-----------------|-----------|------------|-------|-------|
| ENSG00000269028 | *         | 0.450      | 0.284 | 0.564 |
| ENSG00000255823 | *         | 0.399      | 0.237 | 0.512 |
| ENSG00000169710 | FASN*     | 0.298      | 0.263 | 0.321 |
| ENSG00000087460 | GNAS*     | 0.294      | 0.213 | 0.363 |
| ENSG00000096384 | HSP90AB1* | 0.285      | 0.213 | 0.348 |
| ENSG00000141736 | ERBB2*    | 0.278      | 0.137 | 0.326 |
| ENSG00000132475 | H3-3B*    | 0.262      | 0.127 | 0.321 |
| ENSG00000137818 | RPLP1*    | 0.255      | 0.217 | 0.340 |
| ENSG00000177600 | RPLP2*    | 0.242      | 0.148 | 0.301 |
| ENSG00000184009 | ACTG1*    | 0.238      | 0.113 | 0.375 |
| ENSG00000125730 | C3*       | 0.234      | 0.188 | 0.292 |
| ENSG00000254709 | IGLL5*    | 0.226      | 0.050 | 0.433 |
| ENSG00000115414 | FN1*      | 0.223      | 0.052 | 0.373 |
| ENSG00000166710 | B2M*      | 0.221      | 0.138 | 0.340 |
| ENSG00000108821 | COL1A1*   | 0.220      | 0.107 | 0.285 |
| ENSG00000106624 | AEBP1*    | 0.208      | 0.155 | 0.263 |
| ENSG00000187514 | PTMA*     | 0.204      | 0.160 | 0.336 |
| ENSG00000130203 | APOE*     | 0.191      | 0.158 | 0.281 |
| ENSG00000162511 | LAPTM5*   | 0.188      | 0.083 | 0.217 |
| ENSG00000204287 | HLA-DRA*  | 0.186      | 0.130 | 0.307 |
| ENSG00000234745 | HLA-B*    | 0.180      | 0.052 | 0.210 |
| ENSG00000100219 | XBP1*     | 0.177      | 0.064 | 0.255 |
| ENSG00000163399 | ATP1A1*   | 0.172      | 0.095 | 0.246 |
| ENSG00000026025 | VIM*      | 0.169      | 0.136 | 0.234 |
| ENSG00000161960 | EIF4A1*   | 0.166      | 0.117 | 0.256 |
| ENSG00000196576 | PLXNB2*   | 0.162      | 0.027 | 0.214 |
| ENSG00000164733 | CTSB*     | 0.162      | 0.116 | 0.197 |
| ENSG00000180879 | SSR4*     | 0.160      | 0.059 | 0.194 |
| ENSG00000164692 | COL1A2*   | 0.158      | 0.117 | 0.234 |
| ENSG00000182492 | BGN*      | 0.152      | 0.082 | 0.213 |
| ENSG00000149591 | TAGLN*    | 0.146      | 0.128 | 0.256 |
| ENSG00000019582 | CD74*     | 0.141      | 0.109 | 0.203 |
| ENSG00000142173 | COL6A2*   | 0.139      | 0.092 | 0.202 |
| ENSG00000197746 | PSAP*     | 0.137      | 0.039 | 0.181 |
| ENSG00000204525 | HLA-C*    | 0.136      | 0.097 | 0.162 |
| ENSG00000101439 | CST3*     | 0.135      | 0.032 | 0.192 |

continued

| Gene                                         | Symbol   | median $r$ | CI lo  | CI hi |
|----------------------------------------------|----------|------------|--------|-------|
| ENSG00000087086                              | FTL*     | 0.133      | 0.072  | 0.216 |
| ENSG00000251562                              | MALAT1*  | 0.133      | 0.034  | 0.182 |
| ENSG00000173372                              | C1QA*    | 0.130      | 0.057  | 0.211 |
| ENSG00000106211                              | HSPB1*   | 0.127      | 0.093  | 0.148 |
| ENSG00000233276                              | GPX1     | 0.126      | -0.023 | 0.173 |
| ENSG00000142949                              | PTPRF*   | 0.124      | 0.064  | 0.203 |
| ENSG00000197756                              | RPL37A*  | 0.123      | 0.052  | 0.140 |
| ENSG00000070756                              | PABPC1*  | 0.123      | 0.061  | 0.148 |
| ENSG00000075624                              | ACTB*    | 0.120      | 0.057  | 0.154 |
| ENSG00000101335                              | MYL9*    | 0.120      | 0.055  | 0.165 |
| ENSG00000171345                              | KRT19*   | 0.119      | 0.032  | 0.179 |
| ENSG00000205542                              | TMSB4X*  | 0.118      | 0.051  | 0.227 |
| ENSG00000213741                              | RPS29*   | 0.118      | 0.053  | 0.145 |
| ENSG00000206503                              | HLA-A*   | 0.117      | 0.069  | 0.148 |
| ENSG00000231500                              | RPS18*   | 0.115      | 0.056  | 0.188 |
| ENSG00000117450                              | PRDX1*   | 0.113      | 0.062  | 0.158 |
| ENSG00000108107                              | RPL28*   | 0.113      | 0.053  | 0.162 |
| ENSG00000159840                              | ZYX*     | 0.112      | 0.049  | 0.131 |
| ENSG00000089157                              | RPLP0*   | 0.112      | 0.092  | 0.157 |
| ENSG00000170421                              | KRT8*    | 0.112      | 0.075  | 0.170 |
| ENSG00000133112                              | TPT1*    | 0.110      | 0.018  | 0.203 |
| ENSG00000177954                              | RPS27*   | 0.110      | 0.063  | 0.200 |
| ENSG00000130208                              | APOC1*   | 0.109      | 0.052  | 0.149 |
| ENSG00000117984                              | CTSD*    | 0.106      | 0.028  | 0.277 |
| ENSG00000163682                              | RPL9*    | 0.105      | 0.063  | 0.178 |
| ENSG00000142541                              | RPL13A*  | 0.104      | 0.049  | 0.167 |
| ENSG00000164587                              | RPS14*   | 0.100      | 0.072  | 0.182 |
| ENSG00000115884                              | SDC1*    | 0.099      | 0.016  | 0.120 |
| ENSG00000166441                              | RPL27A*  | 0.099      | 0.059  | 0.191 |
| ENSG00000105223                              | PLD3*    | 0.097      | 0.048  | 0.110 |
| ENSG00000111640                              | GAPDH*   | 0.094      | 0.024  | 0.158 |
| __ambiguous[ENSG00000277957+ENSG00000161960] | *        | 0.093      | 0.056  | 0.152 |
| ENSG00000123416                              | TUBA1B*  | 0.092      | 0.034  | 0.260 |
| ENSG00000112306                              | RPS12*   | 0.091      | 0.063  | 0.116 |
| ENSG00000124664                              | SPDEF*   | 0.089      | 0.060  | 0.127 |
| ENSG00000172531                              | PPP1CA*  | 0.087      | 0.066  | 0.121 |
| ENSG00000163191                              | S100A11* | 0.086      | 0.038  | 0.131 |
| ENSG00000279483                              | *        | 0.086      | 0.060  | 0.108 |
| ENSG00000170889                              | RPS9*    | 0.085      | 0.011  | 0.125 |
| __ambiguous[ENSG00000241343+ENSG00000257529] | *        | 0.085      | 0.041  | 0.169 |
| ENSG00000143947                              | RPS27A*  | 0.085      | 0.047  | 0.147 |
| ENSG00000231925                              | TAPBP*   | 0.084      | 0.031  | 0.119 |
| ENSG00000141741                              | MIEN1*   | 0.084      | 0.034  | 0.099 |
| ENSG00000108344                              | PSMD3*   | 0.083      | 0.038  | 0.168 |

continued

| Gene                                          | Symbol    | median $r$ | CI lo  | CI hi |
|-----------------------------------------------|-----------|------------|--------|-------|
| ENSG00000114353                               | GNAI2*    | 0.083      | 0.050  | 0.101 |
| ENSG00000110651                               | CD81*     | 0.083      | 0.024  | 0.104 |
| ENSG00000145425                               | RPS3A*    | 0.083      | 0.025  | 0.105 |
| ENSG00000107223                               | EDF1*     | 0.082      | 0.052  | 0.109 |
| ENSG00000171863                               | RPS7      | 0.081      | -0.011 | 0.133 |
| ENSG00000167526                               | RPL13*    | 0.079      | 0.055  | 0.190 |
| ENSG00000184897                               | H1-10*    | 0.078      | 0.027  | 0.108 |
| ENSG00000034510                               | TMSB10*   | 0.078      | 0.031  | 0.123 |
| ENSG00000109475                               | RPL34*    | 0.078      | 0.056  | 0.169 |
| ENSG00000167995                               | BEST1     | 0.078      | -0.001 | 0.130 |
| ENSG00000182809                               | CRIP2*    | 0.078      | 0.029  | 0.123 |
| ENSG00000221983                               | UBA52*    | 0.077      | 0.049  | 0.122 |
| ENSG00000071082                               | RPL31*    | 0.076      | 0.043  | 0.141 |
| ENSG00000108639                               | SYNGR2*   | 0.076      | 0.038  | 0.139 |
| ...ambiguous[ENSG00000185883+ENSG00000260272] | *         | 0.076      | 0.048  | 0.108 |
| ENSG00000131748                               | STARD3*   | 0.076      | 0.023  | 0.101 |
| ENSG00000186468                               | RPS23*    | 0.076      | 0.025  | 0.130 |
| ENSG00000174748                               | RPL15*    | 0.076      | 0.033  | 0.136 |
| ENSG00000118181                               | RPS25*    | 0.076      | 0.028  | 0.137 |
| ENSG00000105640                               | RPL18A*   | 0.073      | 0.050  | 0.124 |
| ENSG00000102265                               | TIMP1*    | 0.072      | 0.031  | 0.120 |
| ENSG00000124172                               | ATP5F1E*  | 0.070      | 0.034  | 0.195 |
| ENSG00000083845                               | RPS5*     | 0.069      | 0.013  | 0.095 |
| ENSG00000063177                               | RPL18*    | 0.069      | 0.034  | 0.081 |
| ENSG00000130520                               | LSM4*     | 0.069      | 0.025  | 0.122 |
| ENSG00000160789                               | LMNA*     | 0.069      | 0.034  | 0.117 |
| ENSG00000105193                               | RPS16*    | 0.069      | 0.008  | 0.102 |
| ENSG00000058262                               | SEC61A1*  | 0.069      | 0.049  | 0.114 |
| ENSG00000008988                               | RPS20*    | 0.068      | 0.045  | 0.108 |
| ENSG00000113140                               | SPARC*    | 0.068      | 0.023  | 0.113 |
| ENSG00000130255                               | RPL36*    | 0.068      | 0.019  | 0.160 |
| ENSG00000105372                               | RPS19*    | 0.068      | 0.026  | 0.164 |
| ENSG00000196405                               | EVL       | 0.068      | -0.006 | 0.147 |
| ENSG00000138326                               | RPS24*    | 0.067      | 0.034  | 0.151 |
| ENSG00000179218                               | CALR*     | 0.067      | 0.021  | 0.088 |
| ENSG00000108518                               | PFN1*     | 0.067      | 0.039  | 0.110 |
| ENSG00000142669                               | SH3BGRL3* | 0.067      | 0.018  | 0.099 |
| ENSG00000224389                               | C4B*      | 0.065      | 0.031  | 0.132 |
| ENSG00000279274                               | *         | 0.065      | 0.017  | 0.100 |
| ENSG00000025708                               | TYMP*     | 0.065      | 0.041  | 0.100 |
| ENSG00000198034                               | RPS4X     | 0.064      | -0.040 | 0.103 |
| ENSG00000108679                               | LGALS3BP* | 0.064      | 0.030  | 0.093 |
| ENSG00000149806                               | FAU*      | 0.064      | 0.047  | 0.111 |
| ENSG00000233927                               | RPS28*    | 0.064      | 0.012  | 0.087 |

continued

| Gene                                          | Symbol   | median $r$ | CI lo  | CI hi |
|-----------------------------------------------|----------|------------|--------|-------|
| ENSG00000143420                               | ENSA*    | 0.063      | 0.037  | 0.111 |
| ENSG00000182899                               | RPL35A*  | 0.063      | 0.033  | 0.092 |
| ENSG00000145592                               | RPL37*   | 0.063      | 0.025  | 0.117 |
| ENSG00000173369                               | C1QB*    | 0.062      | 0.035  | 0.105 |
| ENSG00000165215                               | CLDN3*   | 0.062      | 0.030  | 0.104 |
| ENSG00000147403                               | RPL10*   | 0.062      | 0.019  | 0.089 |
| ENSG00000164919                               | COX6C*   | 0.061      | 0.039  | 0.122 |
| ENSG00000272196                               | *        | 0.061      | 0.032  | 0.107 |
| ENSG00000142534                               | RPS11*   | 0.061      | 0.008  | 0.100 |
| ENSG00000167468                               | GPX4*    | 0.061      | 0.046  | 0.082 |
| ENSG00000051523                               | CYBA*    | 0.060      | 0.023  | 0.095 |
| ENSG00000137154                               | RPS6*    | 0.060      | 0.031  | 0.123 |
| ENSG00000182774                               | RPS17*   | 0.060      | 0.031  | 0.096 |
| ENSG00000168528                               | SERINC2* | 0.059      | 0.046  | 0.096 |
| ENSG00000173801                               | JUP*     | 0.058      | 0.013  | 0.095 |
| ENSG00000011304                               | PTBP1*   | 0.058      | 0.017  | 0.119 |
| ENSG00000131469                               | RPL27    | 0.057      | -0.008 | 0.083 |
| ENSG00000106153                               | CHCHD2   | 0.057      | -0.008 | 0.146 |
| ENSG00000170315                               | UBB*     | 0.057      | 0.012  | 0.113 |
| ENSG00000126267                               | COX6B1*  | 0.057      | 0.015  | 0.088 |
| ...ambiguous[ENSG00000219200+ENSG00000161939] | *        | 0.056      | 0.030  | 0.094 |
| ENSG00000109062                               | NHERF1*  | 0.056      | 0.014  | 0.111 |
| ENSG00000130303                               | BST2*    | 0.055      | 0.034  | 0.088 |
| ...ambiguous[ENSG00000254772+ENSG00000255508] |          | 0.055      | -0.011 | 0.074 |
| ENSG00000125970                               | RALY*    | 0.055      | 0.009  | 0.138 |
| ENSG00000155366                               | RHOC*    | 0.054      | 0.038  | 0.072 |
| ENSG00000114867                               | EIF4G1*  | 0.054      | 0.022  | 0.074 |
| ENSG00000156482                               | RPL30*   | 0.053      | 0.008  | 0.083 |
| ENSG00000196924                               | FLNA*    | 0.053      | 0.043  | 0.095 |
| ENSG00000142937                               | RPS8     | 0.053      | -0.035 | 0.089 |
| ENSG00000141753                               | IGFBP4   | 0.053      | -0.005 | 0.107 |
| ENSG00000185624                               | P4HB*    | 0.052      | 0.036  | 0.138 |
| ENSG00000104964                               | TLE5*    | 0.052      | 0.030  | 0.064 |
| ENSG00000104529                               | EEF1D    | 0.052      | -0.019 | 0.067 |
| ENSG00000172354                               | GNB2*    | 0.052      | 0.021  | 0.076 |
| ENSG00000147684                               | NDUFB9*  | 0.050      | 0.020  | 0.082 |
| ENSG00000134419                               | RPS15A   | 0.049      | -0.002 | 0.165 |
| ENSG00000132507                               | EIF5A    | 0.048      | -0.013 | 0.070 |
| ENSG00000150991                               | UBC      | 0.047      | -0.013 | 0.089 |
| ENSG00000142676                               | RPL11*   | 0.047      | 0.004  | 0.072 |
| ENSG00000052344                               | PRSS8*   | 0.046      | 0.006  | 0.118 |
| ENSG00000100316                               | RPL3*    | 0.046      | 0.026  | 0.110 |
| ENSG00000100300                               | TSPO     | 0.046      | -0.010 | 0.064 |
| ENSG00000115268                               | RPS15*   | 0.045      | 0.010  | 0.066 |

continued

| Gene                                          | Symbol   | median $r$ | CI lo  | CI hi |
|-----------------------------------------------|----------|------------|--------|-------|
| ENSG00000168542                               | COL3A1   | 0.045      | -0.014 | 0.253 |
| ENSG00000104904                               | OAZ1     | 0.045      | -0.007 | 0.086 |
| ENSG00000174444                               | RPL4*    | 0.044      | 0.031  | 0.082 |
| ENSG00000160932                               | LY6E*    | 0.044      | 0.032  | 0.092 |
| ENSG00000213988                               | ZNF90*   | 0.042      | 0.015  | 0.074 |
| ENSG00000117410                               | ATP6V0B  | 0.041      | -0.015 | 0.070 |
| ENSG00000114391                               | RPL24*   | 0.041      | 0.015  | 0.051 |
| ENSG00000130725                               | UBE2M*   | 0.041      | 0.021  | 0.081 |
| ENSG00000197958                               | RPL12    | 0.039      | -0.014 | 0.072 |
| ...ambiguous[ENSG00000124614+ENSG00000270800] |          | 0.039      | -0.019 | 0.062 |
| ENSG00000147604                               | RPL7     | 0.039      | -0.021 | 0.091 |
| ...ambiguous[ENSG00000257341+ENSG00000213145] | *        | 0.038      | 0.005  | 0.068 |
| ENSG00000172270                               | BSG*     | 0.037      | 0.021  | 0.049 |
| ENSG00000167815                               | PRDX2*   | 0.037      | 0.019  | 0.062 |
| ...ambiguous[ENSG00000251357+ENSG00000240972] | *        | 0.036      | 0.005  | 0.073 |
| ENSG00000167996                               | FTH1     | 0.035      | -0.043 | 0.098 |
| ENSG00000178719                               | GRINA    | 0.034      | -0.001 | 0.086 |
| ENSG00000186010                               | NDUFA13  | 0.034      | -0.031 | 0.049 |
| ENSG00000163479                               | SSR2     | 0.034      | -0.008 | 0.049 |
| ENSG00000135404                               | CD63     | 0.033      | -0.016 | 0.063 |
| ENSG00000092841                               | MYL6     | 0.033      | -0.005 | 0.051 |
| ENSG00000173272                               | MZT2A    | 0.032      | -0.012 | 0.058 |
| ENSG00000162244                               | RPL29    | 0.032      | -0.004 | 0.075 |
| ENSG00000178980                               | SELENOW* | 0.032      | 0.004  | 0.071 |
| ENSG00000105404                               | RABAC1*  | 0.032      | 0.005  | 0.080 |
| ENSG00000172809                               | RPL38*   | 0.031      | 0.015  | 0.069 |
| ENSG00000135480                               | KRT7     | 0.031      | -0.001 | 0.096 |
| ENSG00000140264                               | SERF2    | 0.031      | -0.009 | 0.053 |
| ENSG00000135940                               | COX5B    | 0.031      | -0.016 | 0.071 |
| ENSG00000124614                               | RPS10    | 0.030      | -0.010 | 0.071 |
| ENSG00000149925                               | ALDOA    | 0.030      | -0.010 | 0.065 |
| ENSG00000169223                               | LMAN2    | 0.030      | -0.000 | 0.122 |
| ENSG00000149273                               | RPS3     | 0.030      | -0.010 | 0.090 |
| ENSG00000115457                               | IGFBP2   | 0.029      | -0.010 | 0.095 |
| ENSG00000071553                               | ATP6AP1* | 0.029      | 0.010  | 0.084 |
| ENSG00000143774                               | GUK1     | 0.028      | -0.004 | 0.072 |
| ENSG00000111669                               | TPI1     | 0.028      | -0.022 | 0.071 |
| ENSG00000176046                               | NUPR1    | 0.026      | -0.010 | 0.044 |
| ENSG00000171858                               | RPS21    | 0.025      | -0.015 | 0.050 |
| ENSG00000214530                               | STARD10  | 0.022      | -0.020 | 0.038 |
| ENSG00000140988                               | RPS2     | 0.022      | -0.002 | 0.039 |
| ENSG00000136942                               | RPL35    | 0.022      | -0.035 | 0.116 |
| ENSG00000152082                               | MZT2B    | 0.022      | -0.000 | 0.041 |
| ENSG00000197956                               | S100A6   | 0.021      | -0.023 | 0.063 |

continued

| Gene                                          | Symbol   | median $r$ | CI lo  | CI hi |
|-----------------------------------------------|----------|------------|--------|-------|
| ENSG00000123349                               | PFDN5*   | 0.020      | 0.009  | 0.052 |
| ENSG00000127540                               | UQCR11   | 0.020      | -0.023 | 0.035 |
| ENSG00000204592                               | HLA-E*   | 0.019      | 0.006  | 0.128 |
| ...ambiguous[ENSG00000243678+ENSG00000011052] |          | 0.019      | -0.008 | 0.116 |
| ENSG00000176340                               | COX8A*   | 0.019      | 0.005  | 0.045 |
| ENSG00000188846                               | RPL14    | 0.018      | -0.022 | 0.030 |
| ENSG00000174886                               | NDUFA11* | 0.018      | 0.011  | 0.033 |
| ENSG00000167658                               | EEF2     | 0.018      | -0.015 | 0.051 |
| ENSG00000171222                               | SCAND1   | 0.017      | -0.001 | 0.054 |
| ENSG00000144426                               | NBEAL1   | 0.017      | -0.009 | 0.049 |
| ENSG00000148303                               | RPL7A    | 0.017      | -0.016 | 0.056 |
| ENSG00000169100                               | SLC25A6  | 0.016      | -0.005 | 0.041 |
| ENSG00000145494                               | NDUFS6   | 0.016      | -0.008 | 0.045 |
| ENSG00000111057                               | KRT18    | 0.016      | -0.010 | 0.033 |
| ENSG00000158710                               | TAGLN2*  | 0.016      | 0.002  | 0.049 |
| ...ambiguous[ENSG00000185885+ENSG00000185201] |          | 0.015      | -0.047 | 0.054 |
| ENSG00000030582                               | GRN      | 0.015      | -0.010 | 0.094 |
| ENSG00000277791                               | PSMB3    | 0.014      | -0.030 | 0.084 |
| ENSG00000123989                               | CHPF     | 0.012      | -0.011 | 0.091 |
| ENSG00000131143                               | COX4I1   | 0.011      | -0.016 | 0.022 |
| ENSG00000198242                               | RPL23A   | 0.010      | -0.037 | 0.045 |
| ENSG00000161016                               | RPL8     | 0.010      | -0.012 | 0.085 |
| ENSG00000185825                               | BCAP31   | 0.009      | -0.012 | 0.031 |
| ENSG00000141522                               | ARHGDIA  | 0.008      | -0.015 | 0.038 |
| ENSG00000277972                               | CISD3    | 0.008      | -0.013 | 0.043 |
| ...ambiguous[ENSG00000262526+ENSG00000170296] |          | 0.007      | -0.043 | 0.051 |
| ENSG00000185499                               | MUC1     | 0.005      | -0.076 | 0.029 |
| ENSG00000074800                               | ENO1     | 0.005      | -0.052 | 0.051 |
| ENSG00000172757                               | CFL1     | 0.004      | -0.020 | 0.029 |
| ENSG00000177700                               | POLR2L   | 0.003      | -0.025 | 0.037 |
| ENSG00000178952                               | TUFM     | 0.003      | -0.019 | 0.068 |
| ENSG00000165949                               | IFI27    | 0.001      | -0.019 | 0.017 |
| ENSG00000163220                               | S100A9   | 0.001      | -0.028 | 0.050 |
| ENSG00000187608                               | ISG15    | 0.001      | -0.021 | 0.067 |
| ENSG00000126247                               | CAPNS1   | 0.000      | -0.022 | 0.013 |
| ENSG00000111341                               | MGP      | -0.002     | -0.099 | 0.050 |
| ENSG00000110492                               | MDK      | -0.002     | -0.046 | 0.038 |
| ENSG00000135390                               | ATP5MC2  | -0.002     | -0.056 | 0.038 |
| ...ambiguous[ENSG00000273259+ENSG00000196136] |          | -0.004     | -0.035 | 0.018 |
| ENSG00000140990                               | NDUFB10  | -0.004     | -0.019 | 0.023 |
| ENSG00000103363                               | ELOB     | -0.005     | -0.029 | 0.034 |
| ENSG00000144713                               | RPL32    | -0.006     | -0.047 | 0.023 |
| ENSG00000184840                               | TMED9    | -0.006     | -0.093 | 0.035 |
| ENSG00000143546                               | S100A8   | -0.006     | -0.054 | 0.047 |

continued

| Gene            | Symbol | median $r$ | CI lo  | CI hi  |
|-----------------|--------|------------|--------|--------|
| ENSG00000204628 | RACK1  | -0.008     | -0.020 | 0.046  |
| ENSG00000142089 | IFITM3 | -0.011     | -0.041 | 0.046  |
| ENSG00000160862 | AZGP1  | -0.011     | -0.066 | 0.005  |
| ENSG00000246705 | H2AJ   | -0.012     | -0.040 | 0.039  |
| ENSG00000159335 | PTMS   | -0.020     | -0.030 | 0.027  |
| ENSG00000160180 | TFF3   | -0.023     | -0.079 | 0.013  |
| ENSG00000125691 | RPL23  | -0.043     | -0.099 | 0.004  |
| ENSG00000110092 | CCND1  | -0.046     | -0.119 | 0.054  |
| ENSG00000125534 | PPDPF  | -0.050     | -0.071 | 0.007  |
| ENSG00000108298 | RPL19  | -0.065     | -0.137 | -0.012 |

## C Locked formulas index

1. Pearson  $r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$
2. Spearman  $\rho$  (Pearson on ranks)
3. Ridge  $\hat{\beta} = (X^\top X + \lambda P)^{-1} X^\top Y$
4. Gaussian kernel  $W_{ij} = \exp(-d_{ij}^2/2\ell^2)$ , row-normalized
5. Spatial smoothing  $\hat{y}^{(2)} = (1 - \alpha)\hat{y} + \alpha W \hat{y}$
6. Moran's  $I$  (see Methods)
7. Beer-Lambert  $OD = -\ln(I/I_0)$ ; deconvolution  $C = ODM^{-1}$
8. GLCM contrast / homogeneity / energy / correlation (see Methods)
9. Shannon entropy  $H = -\sum p \log_2 p$
10. AUC via Mann-Whitney  $U/(n_1 n_0)$
11. Benjamini-Hochberg adjusted  $p_{(i)} \cdot m/i$
12. Permutation  $p = (1 + \#\{|null| \geq |obs|\})/(1 + n)$
13. Wilcoxon signed-rank statistic
14. Brier score  $\frac{1}{n} \sum (p_i - y_i)^2$
15. Softmax  $\alpha_i = e^{s_i/\tau} / \sum_j e^{s_j/\tau}$
16. Library normalization  $\log(1 + 10^4 x / \sum x)$

**Provenance note.** The appendix tables from here on (per-fold, transport, scanner coefficients, top predictable genes, enrichment, ledgers, cohort manifest) are Phase-1 artifacts from the handcrafted-feature pipeline (commit 84b076a), preserved for provenance. Phase-2 equivalents are Appendix 7–8 and the refreshed scanner outputs in `results/`.

## D Per-fold results

| Patient | sections | $\lambda$ | $\alpha$ | M1 median $r$ | M2 median $r$ |
|---------|----------|-----------|----------|---------------|---------------|
| BC23287 | 3        | 100.0     | 0.5      | -0.007        | -0.010        |
| BC23450 | 3        | 100.0     | 0.4      | 0.016         | 0.019         |
| BC23567 | 3        | 100.0     | 0.5      | 0.009         | 0.009         |
| BC23810 | 3        | 100.0     | 0.5      | 0.011         | 0.014         |
| BC23901 | 2        | 100.0     | 0.5      | 0.009         | 0.013         |
| BC23903 | 3        | 100.0     | 0.5      | 0.043         | 0.055         |
| BC23944 | 3        | 100.0     | 0.5      | 0.029         | 0.033         |
| BC24044 | 3        | 100.0     | 0.5      | 0.040         | 0.049         |
| BC24105 | 3        | 100.0     | 0.5      | -0.017        | -0.022        |
| BC24220 | 3        | 100.0     | 0.5      | 0.025         | 0.034         |

## E Transport results

| Subtype   | Section    | median per-gene $r$ |
|-----------|------------|---------------------|
| Luminal_A | BC23268_C1 | 0.072               |
| Luminal_A | BC23268_C2 | 0.061               |
| Luminal_A | BC23268_D1 | 0.076               |
| Luminal_A | BC23269_C1 | 0.040               |
| Luminal_A | BC23269_C2 | 0.025               |
| Luminal_A | BC23269_D1 | 0.031               |
| Luminal_A | BC23272_D2 | 0.050               |
| Luminal_A | BC23272_E1 | 0.075               |
| Luminal_A | BC23272_E2 | 0.021               |
| Luminal_A | BC24223_D2 | -0.024              |
| Luminal_A | BC24223_E1 | -0.013              |
| Luminal_A | BC24223_E2 | 0.000               |
| Luminal_B | BC23270_D2 | -0.007              |
| Luminal_B | BC23270_E1 | 0.041               |
| Luminal_B | BC23270_E2 | 0.023               |
| Luminal_B | BC23277_D2 | 0.015               |
| Luminal_B | BC23277_E1 | 0.027               |
| Luminal_B | BC23277_E2 | -0.025              |
| Luminal_B | BC23506_C1 | 0.016               |
| Luminal_B | BC23506_C2 | -0.004              |
| Luminal_B | BC23506_D1 | -0.001              |
| Luminal_B | BC23508_D2 | 0.037               |
| Luminal_B | BC23508_E1 | 0.026               |
| Luminal_B | BC23508_E2 | 0.043               |
| Luminal_B | BC23895_C1 | 0.013               |
| Luminal_B | BC23895_C2 | 0.014               |

| Subtype   | Section    | median $r$ |
|-----------|------------|------------|
| Luminal_B | BC23895_D1 | 0.005      |
| TNBC      | BC23209_C1 | 0.004      |
| TNBC      | BC23209_C2 | 0.004      |
| TNBC      | BC23209_D1 | -0.024     |
| TNBC      | BC23288_D2 | 0.021      |
| TNBC      | BC23288_E1 | 0.051      |
| TNBC      | BC23288_E2 | 0.029      |
| TNBC      | BC23377_C1 | 0.010      |
| TNBC      | BC23377_C2 | -0.005     |
| TNBC      | BC23377_D1 | -0.007     |
| TNBC      | BC23803_D2 | 0.006      |
| TNBC      | BC23803_E1 | 0.038      |
| TNBC      | BC23803_E2 | 0.019      |

## F Scanner coefficients

| Feature        | coefficient | $z$    | $p$      |
|----------------|-------------|--------|----------|
| glcm0_energy   | 24.566      | 5.52   | 3.41e-08 |
| glcm1_energy   | -23.145     | -5.13  | 2.90e-07 |
| glcm3_energy   | 12.963      | 3.56   | 3.67e-04 |
| glcm2_energy   | -12.931     | -3.60  | 3.23e-04 |
| b_mean         | 11.592      | 18.82  | 5.46e-79 |
| r_mean         | -5.979      | -19.25 | 1.51e-82 |
| eosin_mean     | 5.229       | 12.23  | 2.26e-34 |
| dab_mean       | 4.647       | 15.01  | 6.58e-51 |
| r_std          | 2.601       | 12.53  | 5.15e-36 |
| g_mean         | -2.417      | -7.49  | 6.93e-14 |
| hema_mean      | 2.409       | 10.49  | 9.52e-26 |
| hema_std       | -2.103      | -17.03 | 4.88e-65 |
| eosin_std      | 2.020       | 12.40  | 2.58e-35 |
| b_std          | -1.372      | -8.12  | 4.63e-16 |
| glcm2_homog    | -1.319      | -1.50  | 1.33e-01 |
| g_std          | -1.137      | -5.40  | 6.84e-08 |
| glcm1_homog    | -0.915      | -1.10  | 2.69e-01 |
| gray_entropy   | -0.745      | -9.48  | 2.61e-21 |
| glcm3_homog    | 0.649       | 0.75   | 4.55e-01 |
| glcm0_homog    | -0.617      | -0.74  | 4.60e-01 |
| glcm1_contrast | 0.528       | 0.96   | 3.35e-01 |
| r_skew         | -0.515      | -3.55  | 3.84e-04 |
| hema_entropy   | 0.422       | 11.23  | 2.77e-29 |
| glcm3_corr     | 0.387       | 4.08   | 4.59e-05 |
| glcm1_corr     | -0.365      | -4.66  | 3.18e-06 |
| grad_mag_mean  | -0.365      | -2.00  | 4.55e-02 |

| Feature         | coef   | $z$   | $p$      |
|-----------------|--------|-------|----------|
| dab_std         | 0.279  | 3.92  | 8.80e-05 |
| glcm2_contrast  | -0.247 | -0.43 | 6.66e-01 |
| g_skew          | 0.236  | 1.53  | 1.25e-01 |
| glcm0_corr      | -0.213 | -2.69 | 7.05e-03 |
| glcm0_contrast  | 0.189  | 0.34  | 7.34e-01 |
| glcm2_corr      | -0.135 | -1.44 | 1.51e-01 |
| hema_dense_frac | 0.133  | 5.05  | 4.52e-07 |
| r_kurt          | -0.114 | -0.76 | 4.48e-01 |
| hema_fracpos    | 0.078  | 3.38  | 7.22e-04 |
| b_kurt          | -0.065 | -0.42 | 6.77e-01 |
| dab_fracpos     | 0.045  | 2.01  | 4.47e-02 |
| glcm3_contrast  | 0.019  | 0.03  | 9.74e-01 |
| b_skew          | 0.015  | 0.10  | 9.19e-01 |
| eosin_fracpos   | -0.006 | -0.24 | 8.09e-01 |
| g_kurt          | 0.001  | 0.01  | 9.95e-01 |

## G Top predictable genes

| Gene (Ensembl)  | M1 median $r$ | M2 median $r$ |
|-----------------|---------------|---------------|
| ENSG00000177600 | 0.144         | 0.154         |
| ENSG00000096384 | 0.140         | 0.138         |
| ENSG00000141738 | 0.115         | 0.140         |
| ENSG00000137818 | 0.111         | 0.144         |
| ENSG00000269028 | 0.110         | 0.153         |
| ENSG00000108821 | 0.109         | 0.137         |
| ENSG00000132475 | 0.105         | 0.140         |
| ENSG00000255823 | 0.104         | 0.122         |
| ENSG00000166710 | 0.098         | 0.114         |
| ENSG00000087086 | 0.090         | 0.104         |
| ENSG00000196576 | 0.087         | 0.089         |
| ENSG00000279483 | 0.087         | 0.104         |
| ENSG00000130203 | 0.086         | 0.120         |
| ENSG00000161960 | 0.078         | 0.116         |
| ENSG00000168528 | 0.076         | 0.095         |
| ENSG00000115414 | 0.075         | 0.082         |
| ENSG00000169710 | 0.073         | 0.090         |
| ENSG00000141736 | 0.068         | 0.077         |
| ENSG00000087460 | 0.068         | 0.092         |
| ENSG00000234745 | 0.067         | 0.080         |
| ENSG00000058262 | 0.064         | 0.082         |
| ENSG00000102265 | 0.063         | 0.054         |
| ENSG00000204287 | 0.061         | 0.079         |
| ENSG00000177954 | 0.061         | 0.070         |

| Gene                                          | M1    | M2     |
|-----------------------------------------------|-------|--------|
| ENSG00000075624                               | 0.061 | 0.085  |
| ENSG00000108344                               | 0.060 | 0.052  |
| ENSG00000170421                               | 0.059 | 0.061  |
| ENSG00000089157                               | 0.056 | 0.058  |
| ENSG00000108518                               | 0.055 | 0.070  |
| ENSG00000132507                               | 0.055 | 0.061  |
| ENSG00000130255                               | 0.054 | 0.073  |
| ENSG00000205542                               | 0.053 | 0.069  |
| ENSG00000164692                               | 0.053 | 0.065  |
| ENSG00000197746                               | 0.053 | 0.069  |
| ENSG00000026025                               | 0.052 | 0.069  |
| ENSG00000083845                               | 0.051 | 0.051  |
| ENSG00000100097                               | 0.051 | 0.055  |
| ENSG00000182899                               | 0.050 | 0.050  |
| ENSG00000167526                               | 0.049 | 0.045  |
| ENSG00000142937                               | 0.048 | 0.046  |
| ENSG00000130725                               | 0.048 | 0.047  |
| ENSG00000163399                               | 0.047 | 0.069  |
| ENSG00000164733                               | 0.047 | 0.064  |
| ENSG00000197756                               | 0.046 | 0.060  |
| ENSG00000100219                               | 0.046 | 0.047  |
| ENSG00000108107                               | 0.046 | 0.050  |
| ENSG00000108639                               | 0.046 | 0.038  |
| ENSG00000105223                               | 0.046 | 0.052  |
| __ambiguous [ENSG00000277957+ENSG00000161960] | 0.045 | 0.071  |
| ENSG00000134419                               | 0.044 | 0.065  |
| ENSG00000019582                               | 0.043 | 0.038  |
| ENSG00000149806                               | 0.043 | 0.037  |
| ENSG00000159840                               | 0.043 | -0.006 |
| ENSG00000142949                               | 0.043 | 0.053  |
| ENSG00000198034                               | 0.042 | 0.031  |
| ENSG00000184009                               | 0.041 | 0.042  |
| ENSG00000105193                               | 0.041 | 0.047  |
| __ambiguous [ENSG00000185883+ENSG00000260272] | 0.041 | 0.050  |
| ENSG00000125970                               | 0.040 | 0.069  |
| ENSG00000164587                               | 0.040 | 0.049  |
| ENSG00000158710                               | 0.040 | 0.031  |
| ENSG00000125730                               | 0.039 | 0.043  |
| ENSG00000123416                               | 0.038 | 0.044  |
| ENSG00000173369                               | 0.038 | 0.034  |
| ENSG00000142173                               | 0.037 | 0.049  |
| ENSG00000114867                               | 0.037 | 0.045  |
| ENSG00000173272                               | 0.035 | 0.023  |
| ENSG00000117450                               | 0.035 | 0.062  |
| ENSG00000279274                               | 0.035 | 0.054  |

| Gene                                          | M1    | M2    |
|-----------------------------------------------|-------|-------|
| ENSG00000229117                               | 0.035 | 0.026 |
| ENSG00000231500                               | 0.034 | 0.047 |
| ENSG00000130520                               | 0.034 | 0.031 |
| ENSG00000106153                               | 0.034 | 0.023 |
| ENSG00000142541                               | 0.034 | 0.041 |
| ENSG00000163191                               | 0.034 | 0.051 |
| ENSG00000231925                               | 0.033 | 0.037 |
| ENSG00000145425                               | 0.033 | 0.050 |
| ENSG00000104904                               | 0.033 | 0.038 |
| ENSG00000131748                               | 0.033 | 0.056 |
| ENSG00000105404                               | 0.032 | 0.028 |
| __ambiguous [ENSG00000251357+ENSG00000240972] | 0.032 | 0.024 |
| ENSG00000196405                               | 0.032 | 0.032 |
| ENSG00000126267                               | 0.031 | 0.039 |
| ENSG00000174444                               | 0.030 | 0.023 |
| ENSG00000185624                               | 0.030 | 0.028 |
| ENSG00000118181                               | 0.029 | 0.030 |
| ENSG00000111640                               | 0.029 | 0.033 |
| ENSG00000187514                               | 0.028 | 0.056 |
| ENSG00000170315                               | 0.028 | 0.015 |
| ENSG00000224389                               | 0.028 | 0.019 |
| ENSG00000233927                               | 0.028 | 0.022 |
| ENSG00000143420                               | 0.027 | 0.037 |
| ENSG00000213741                               | 0.027 | 0.023 |
| ENSG00000051523                               | 0.027 | 0.008 |
| ENSG00000277791                               | 0.027 | 0.042 |
| ENSG00000071082                               | 0.027 | 0.051 |
| ENSG00000128322                               | 0.027 | 0.003 |
| ENSG00000071553                               | 0.026 | 0.038 |
| ENSG00000277972                               | 0.026 | 0.016 |
| ENSG00000163682                               | 0.025 | 0.043 |
| ENSG00000109062                               | 0.025 | 0.040 |
| ENSG00000272196                               | 0.025 | 0.025 |
| ENSG00000034510                               | 0.024 | 0.014 |
| ENSG00000123989                               | 0.024 | 0.020 |
| ENSG00000113140                               | 0.024 | 0.022 |
| __ambiguous [ENSG00000243678+ENSG00000011052] | 0.023 | 0.029 |
| ENSG00000171863                               | 0.023 | 0.042 |
| ENSG00000140264                               | 0.023 | 0.034 |
| ENSG00000167468                               | 0.022 | 0.026 |
| ENSG00000178952                               | 0.022 | 0.017 |
| __ambiguous [ENSG00000241343+ENSG00000257529] | 0.022 | 0.024 |
| ENSG00000204525                               | 0.022 | 0.056 |
| ENSG00000141741                               | 0.022 | 0.010 |
| ENSG00000170889                               | 0.021 | 0.019 |

| Gene            | M1    | M2    |
|-----------------|-------|-------|
| ENSG00000141522 | 0.021 | 0.025 |
| ENSG00000142676 | 0.021 | 0.015 |
| ENSG00000140988 | 0.021 | 0.020 |
| ENSG00000110651 | 0.020 | 0.026 |
| ENSG00000115457 | 0.020 | 0.004 |
| ENSG00000162244 | 0.020 | 0.015 |

## H Dataset ledger composition

The 629-record accession ledger summarized by source, with example accessions; the full ledger is `results/accession_ledger.csv`.

| Source                 | Records | Level            | Example accessions                                                                                                                 |
|------------------------|---------|------------------|------------------------------------------------------------------------------------------------------------------------------------|
| CELLxGENE              | 40      | metadata-mined   | cxg:af893e86-8e9f-41f1-a474-ef05359b1fb7;<br>cxg:3a5dbf8a-9b3e-4309-b4c5-d8a024f83734;<br>cxg:16876983-d454-43db-a8ec-77fafca9bf38 |
| GDC                    | 80      | metadata-mined   | TCGA-LUAD; ALCHEMIST-ALCH; MATCH-S1                                                                                                |
| GEO-GSE                | 400     | metadata-mined   | GSE344506; GSE348445; GSE337859                                                                                                    |
| Mendeley-29ntw7sh4r-v5 | 69      | analyzed-primary | BC23287_C1; BC23287_C2; BC23287_D1                                                                                                 |
| TCIA                   | 40      | metadata-mined   | tcia:4D-Lung; tcia:A091105; tcia:ACNS0332                                                                                          |
| Total                  | 629     |                  |                                                                                                                                    |

## I Cohort manifest

One row per analyzed section. Tumor fraction is the share of spots under the pathologist tumor annotation for that section.

| Patient | Section    | Subtype      | Spots | Tumor frac. | HER2+ |
|---------|------------|--------------|-------|-------------|-------|
| BC23287 | BC23287_C1 | HER2_luminal | 256   | 0.00        | yes   |
| BC23287 | BC23287_C2 | HER2_luminal | 334   | 0.00        | yes   |
| BC23287 | BC23287_D1 | HER2_luminal | 357   | 0.00        | yes   |
| BC23450 | BC23450_D2 | HER2_luminal | 316   | 0.00        | yes   |
| BC23450 | BC23450_E1 | HER2_luminal | 301   | 0.00        | yes   |
| BC23450 | BC23450_E2 | HER2_luminal | 318   | 0.00        | yes   |
| BC23901 | BC23901_C2 | HER2_luminal | 292   | 0.00        | yes   |
| BC23901 | BC23901_D1 | HER2_luminal | 297   | 0.00        | yes   |
| BC23944 | BC23944_D2 | HER2_luminal | 414   | 0.00        | yes   |
| BC23944 | BC23944_E1 | HER2_luminal | 423   | 0.00        | yes   |
| BC23944 | BC23944_E2 | HER2_luminal | 403   | 0.00        | yes   |
| BC24220 | BC24220_D2 | HER2_luminal | 432   | 0.00        | yes   |
| BC24220 | BC24220_E1 | HER2_luminal | 428   | 0.00        | yes   |

| Patient | Section    | Subtype          | Spots | Tumor | HER2+ |
|---------|------------|------------------|-------|-------|-------|
| BC24220 | BC24220_E2 | HER2_luminal     | 437   | 0.00  | yes   |
| BC23567 | BC23567_D2 | HER2_non_luminal | 587   | 0.00  | yes   |
| BC23567 | BC23567_E1 | HER2_non_luminal | 572   | 0.00  | yes   |
| BC23567 | BC23567_E2 | HER2_non_luminal | 570   | 0.00  | yes   |
| BC23810 | BC23810_D2 | HER2_non_luminal | 691   | 0.00  | yes   |
| BC23810 | BC23810_E1 | HER2_non_luminal | 695   | 0.00  | yes   |
| BC23810 | BC23810_E2 | HER2_non_luminal | 713   | 0.00  | yes   |
| BC23903 | BC23903_C1 | HER2_non_luminal | 441   | 0.00  | yes   |
| BC23903 | BC23903_C2 | HER2_non_luminal | 468   | 0.00  | yes   |
| BC23903 | BC23903_D1 | HER2_non_luminal | 464   | 0.00  | yes   |
| BC24044 | BC24044_D2 | HER2_non_luminal | 614   | 0.00  | yes   |
| BC24044 | BC24044_E1 | HER2_non_luminal | 603   | 0.00  | yes   |
| BC24044 | BC24044_E2 | HER2_non_luminal | 511   | 0.00  | yes   |
| BC24105 | BC24105_C1 | HER2_non_luminal | 325   | 0.00  | yes   |
| BC24105 | BC24105_C2 | HER2_non_luminal | 297   | 0.00  | yes   |
| BC24105 | BC24105_D1 | HER2_non_luminal | 338   | 0.00  | yes   |
| BC23268 | BC23268_C1 | Luminal_A        | 435   | 0.00  | no    |
| BC23268 | BC23268_C2 | Luminal_A        | 498   | 0.00  | no    |
| BC23268 | BC23268_D1 | Luminal_A        | 492   | 0.00  | no    |
| BC23269 | BC23269_C1 | Luminal_A        | 434   | 0.00  | no    |
| BC23269 | BC23269_C2 | Luminal_A        | 435   | 0.00  | no    |
| BC23269 | BC23269_D1 | Luminal_A        | 433   | 0.00  | no    |
| BC23272 | BC23272_D2 | Luminal_A        | 565   | 0.00  | no    |
| BC23272 | BC23272_E1 | Luminal_A        | 578   | 0.00  | no    |
| BC23272 | BC23272_E2 | Luminal_A        | 548   | 0.00  | no    |
| BC24223 | BC24223_D2 | Luminal_A        | 368   | 0.00  | no    |
| BC24223 | BC24223_E1 | Luminal_A        | 362   | 0.00  | no    |
| BC24223 | BC24223_E2 | Luminal_A        | 363   | 0.00  | no    |
| BC23270 | BC23270_D2 | Luminal_B        | 284   | 0.00  | no    |
| BC23270 | BC23270_E1 | Luminal_B        | 283   | 0.00  | no    |
| BC23270 | BC23270_E2 | Luminal_B        | 284   | 0.00  | no    |
| BC23277 | BC23277_D2 | Luminal_B        | 506   | 0.00  | no    |
| BC23277 | BC23277_E1 | Luminal_B        | 513   | 0.00  | no    |
| BC23277 | BC23277_E2 | Luminal_B        | 566   | 0.00  | no    |
| BC23506 | BC23506_C1 | Luminal_B        | 489   | 0.00  | no    |
| BC23506 | BC23506_C2 | Luminal_B        | 502   | 0.00  | no    |
| BC23506 | BC23506_D1 | Luminal_B        | 482   | 0.00  | no    |
| BC23508 | BC23508_D2 | Luminal_B        | 478   | 0.00  | no    |
| BC23508 | BC23508_E1 | Luminal_B        | 473   | 0.00  | no    |
| BC23508 | BC23508_E2 | Luminal_B        | 504   | 0.00  | no    |
| BC23895 | BC23895_C1 | Luminal_B        | 498   | 0.00  | no    |
| BC23895 | BC23895_C2 | Luminal_B        | 454   | 0.00  | no    |
| BC23895 | BC23895_D1 | Luminal_B        | 518   | 0.00  | no    |
| BC23209 | BC23209_C1 | TNBC             | 294   | 0.00  | no    |
| BC23209 | BC23209_C2 | TNBC             | 318   | 0.00  | no    |

| Patient | Section    | Subtype | Spots | Tumor | HER2+ |
|---------|------------|---------|-------|-------|-------|
| BC23209 | BC23209_D1 | TNBC    | 331   | 0.00  | no    |
| BC23288 | BC23288_D2 | TNBC    | 464   | 0.00  | no    |
| BC23288 | BC23288_E1 | TNBC    | 480   | 0.00  | no    |
| BC23288 | BC23288_E2 | TNBC    | 478   | 0.00  | no    |
| BC23377 | BC23377_C1 | TNBC    | 597   | 0.00  | no    |
| BC23377 | BC23377_C2 | TNBC    | 687   | 0.00  | no    |
| BC23377 | BC23377_D1 | TNBC    | 601   | 0.00  | no    |
| BC23803 | BC23803_D2 | TNBC    | 402   | 0.00  | no    |
| BC23803 | BC23803_E1 | TNBC    | 340   | 0.00  | no    |
| BC23803 | BC23803_E2 | TNBC    | 421   | 0.00  | no    |

## J Pathway enrichment of the top-25 predictable genes

g:Profiler g:GOST (GO:BP, KEGG, Reactome) on the 25 genes with the highest held-out M1 correlation; full response in results/enrichment\_top25.json.

| Source | Term                                                              | <i>p</i> | Genes |
|--------|-------------------------------------------------------------------|----------|-------|
| KEGG   | Antigen processing and presentation                               | 9.22e-04 | 23    |
| REAC   | Scavenging by Class A Receptors                                   | 1.82e-03 | 23    |
| GO:BP  | antigen processing and presentation of endogenous peptide antigen | 3.12e-03 | 23    |
| REAC   | GRB7 events in ERBB2 signaling                                    | 6.68e-03 | 23    |
| GO:BP  | antigen processing and presentation of endogenous antigen         | 6.90e-03 | 23    |
| GO:BP  | wound healing                                                     | 7.13e-03 | 23    |
| GO:BP  | protein biosynthetic process                                      | 8.61e-03 | 23    |
| GO:BP  | translation                                                       | 8.61e-03 | 23    |
| REAC   | Disease                                                           | 9.90e-03 | 23    |
| KEGG   | Vibrio cholerae infection                                         | 1.04e-02 | 23    |
| GO:BP  | positive regulation of T cell mediated cytotoxicity               | 1.19e-02 | 23    |
| REAC   | Axon guidance                                                     | 1.37e-02 | 23    |
| GO:BP  | regulation of cell adhesion                                       | 1.57e-02 | 23    |
| GO:BP  | regulation of developmental process                               | 1.61e-02 | 23    |
| KEGG   | Phagosome                                                         | 1.68e-02 | 23    |

## K Test inventory

| Test module | Tests |
|-------------|-------|
| features    | 11    |
| metrics     | 21    |
| Total       | 32    |

The 32 unit tests pin every metric against known values (perfect/anti correlation, constant-input degeneracies, ridge recovery and shrinkage, kernel normalization, smoothing identity at  $\alpha = 0$ , Moran's  $I$  sign

on clustered vs alternating patterns, AUC 0/0.5/1 anchors, BH monotonicity, permutation- $p$  bounds, Brier anchors, softmax normalization and temperature behavior) and every feature extractor against physical expectations (white-field OD  $\approx 0$ , constant-image GLCM energy 1 and contrast 0, uniform-histogram entropy maximum, feature-vector finiteness and name uniqueness, textured-vs-flat discriminability).

## L Phase-2 scanner: per-section performance

Table 17: Phase-2 scanner: per-section patient-held-out AUC. Sections with undefined AUC (tumor fraction 1.0: BC24105\_C1, BC24105\_C2) are excluded; BC24105\_D1 is retained and reported as a named outlier.

| Patient | Section    | $n$ spots | Tumor frac. | AUC   |
|---------|------------|-----------|-------------|-------|
| BC23287 | BC23287_C1 | 256       | 0.312       | 0.992 |
| BC23287 | BC23287_C2 | 334       | 0.266       | 0.990 |
| BC23287 | BC23287_D1 | 357       | 0.258       | 0.667 |
| BC23450 | BC23450_D2 | 316       | 0.883       | 0.885 |
| BC23450 | BC23450_E1 | 301       | 0.934       | 0.887 |
| BC23450 | BC23450_E2 | 318       | 0.909       | 0.912 |
| BC23567 | BC23567_D2 | 587       | 0.472       | 0.845 |
| BC23567 | BC23567_E1 | 572       | 0.497       | 0.841 |
| BC23567 | BC23567_E2 | 570       | 0.488       | 0.845 |
| BC23810 | BC23810_D2 | 691       | 0.889       | 0.926 |
| BC23810 | BC23810_E1 | 695       | 0.883       | 0.952 |
| BC23810 | BC23810_E2 | 713       | 0.871       | 0.936 |
| BC23901 | BC23901_C2 | 292       | 0.442       | 0.826 |
| BC23901 | BC23901_D1 | 297       | 0.529       | 0.849 |
| BC23903 | BC23903_C1 | 441       | 0.213       | 0.921 |
| BC23903 | BC23903_C2 | 468       | 0.171       | 0.917 |
| BC23903 | BC23903_D1 | 464       | 0.213       | 0.932 |
| BC23944 | BC23944_D2 | 414       | 0.836       | 0.976 |
| BC23944 | BC23944_E1 | 423       | 0.823       | 0.980 |
| BC23944 | BC23944_E2 | 403       | 0.851       | 0.993 |
| BC24044 | BC24044_D2 | 614       | 0.135       | 0.973 |
| BC24044 | BC24044_E1 | 603       | 0.134       | 0.963 |
| BC24044 | BC24044_E2 | 511       | 0.094       | 0.972 |
| BC24105 | BC24105_C1 | 325       | 1.000       | —     |
| BC24105 | BC24105_C2 | 297       | 1.000       | —     |
| BC24105 | BC24105_D1 | 338       | 0.997       | 0.065 |
| BC24220 | BC24220_D2 | 432       | 0.884       | 0.924 |
| BC24220 | BC24220_E1 | 428       | 0.930       | 0.954 |
| BC24220 | BC24220_E2 | 437       | 0.911       | 0.941 |

## M Phase-2 scanner: subgroup analyses

Table 18: Phase-2 scanner subgroup performance (27 evaluable sections).

| Subgroup         | <i>n</i> | Median AUC | IQR            | Range          |
|------------------|----------|------------|----------------|----------------|
| HER2 luminal     | 14       | 0.933      | [0.885, 0.979] | [0.667, 0.993] |
| HER2 non luminal | 13       | 0.926      | [0.845, 0.952] | [0.065, 0.973] |
| low              | 9        | 0.963      | [0.921, 0.973] | [0.667, 0.992] |
| mid              | 9        | 0.849      | [0.845, 0.976] | [0.826, 0.993] |
| high             | 9        | 0.924      | [0.887, 0.941] | [0.065, 0.954] |

## N Phase-2 scanner: coefficient stability

Table 19: Phase-2 scanner coefficient stability across the 10 LOPO folds (768-dimensional embedding-logistic coefficient vectors).

| Metric                              | Value          |
|-------------------------------------|----------------|
| Pairwise Spearman, median           | 0.857          |
| Pairwise Spearman, bootstrap 95% CI | [0.839, 0.874] |
| Pairwise Spearman, range            | [0.753, 0.942] |
| Top-50 feature Jaccard, median      | 0.620          |
| Top-50 sign consistency, median     | 1.000          |

## O Phase-2 scanner: label-permutation controls

Table 20: Phase-2 scanner within-section label-permutation control (20 permutations per section): true AUC vs the permutation null. Exceedances counts permuted fits reaching the true AUC.

| Section    | True AUC | Null median | Null p95 | Exceedances |
|------------|----------|-------------|----------|-------------|
| BC23287_C1 | 0.992    | 0.504       | 0.565    | 0/20        |
| BC23287_C2 | 0.990    | 0.507       | 0.526    | 0/20        |
| BC23287_D1 | 0.667    | 0.515       | 0.575    | 0/20        |
| BC23450_D2 | 0.885    | 0.474       | 0.556    | 0/20        |
| BC23450_E1 | 0.887    | 0.503       | 0.591    | 0/20        |
| BC23450_E2 | 0.912    | 0.509       | 0.595    | 0/20        |
| BC23567_D2 | 0.845    | 0.497       | 0.515    | 0/20        |
| BC23567_E1 | 0.841    | 0.493       | 0.532    | 0/20        |
| BC23567_E2 | 0.845    | 0.497       | 0.532    | 0/20        |
| BC23810_D2 | 0.926    | 0.509       | 0.563    | 0/20        |
| BC23810_E1 | 0.952    | 0.514       | 0.563    | 0/20        |
| BC23810_E2 | 0.936    | 0.490       | 0.537    | 0/20        |
| BC23901_C2 | 0.826    | 0.519       | 0.555    | 0/20        |
| BC23901_D1 | 0.849    | 0.510       | 0.562    | 0/20        |
| BC23903_C1 | 0.921    | 0.506       | 0.557    | 0/20        |

| Section    | True AUC | Null median | Null p95 | Exceedances |
|------------|----------|-------------|----------|-------------|
| BC23903_C2 | 0.917    | 0.504       | 0.547    | 0/20        |
| BC23903_D1 | 0.932    | 0.514       | 0.541    | 0/20        |
| BC23944_D2 | 0.976    | 0.507       | 0.558    | 0/20        |
| BC23944_E1 | 0.980    | 0.498       | 0.533    | 0/20        |
| BC23944_E2 | 0.993    | 0.518       | 0.569    | 0/20        |
| BC24044_D2 | 0.973    | 0.500       | 0.559    | 0/20        |
| BC24044_E1 | 0.963    | 0.503       | 0.570    | 0/20        |
| BC24044_E2 | 0.972    | 0.514       | 0.578    | 0/20        |
| BC24105_D1 | 0.065    | 0.585       | 0.899    | 18/20       |
| BC24220_D2 | 0.924    | 0.500       | 0.574    | 0/20        |
| BC24220_E1 | 0.954    | 0.508       | 0.575    | 0/20        |
| BC24220_E2 | 0.941    | 0.524       | 0.591    | 0/20        |

## P Multi-task per-fold results

Table 21: Multi-task legs per held-out patient (median per-gene Pearson over the 250 train-selected genes). ridge = committed independent ridge; refold = same ridge refit by the multi-task harness (sanity); PLS = partial least squares (25 components); RRR = reduced-rank ridge (rank 25); enet = multi-output elastic net (alpha 1e-3, l1-ratio 0.5, max 500 iterations, random selection, seed 20260929; all folds hit the iteration cap - non-convergence declared).

| patient | ridge  | refold | PLS    | RRR    | enet   | enet n_iter |
|---------|--------|--------|--------|--------|--------|-------------|
| BC23287 | 0.0588 | 0.0520 | 0.0896 | 0.0721 | 0.0467 | 500         |
| BC23450 | 0.0388 | 0.0484 | 0.0734 | 0.0562 | 0.0439 | 500         |
| BC23567 | 0.0334 | 0.0404 | 0.0491 | 0.0500 | 0.0386 | 500         |
| BC23810 | 0.0623 | 0.0720 | 0.0786 | 0.0766 | 0.0658 | 500         |
| BC23901 | 0.0417 | 0.0314 | 0.0494 | 0.0390 | 0.0277 | 500         |
| BC23903 | 0.1063 | 0.1006 | 0.1077 | 0.1041 | 0.0948 | 500         |
| BC23944 | 0.0390 | 0.0323 | 0.0442 | 0.0302 | 0.0268 | 500         |
| BC24044 | 0.0853 | 0.0846 | 0.0966 | 0.0910 | 0.0797 | 500         |
| BC24105 | 0.0312 | 0.0315 | 0.0400 | 0.0366 | 0.0279 | 500         |
| BC24220 | 0.0507 | 0.0542 | 0.0611 | 0.0630 | 0.0511 | 500         |

## Q Phase-2 per-fold results

Per-patient held-out summaries for the Phase-2 ridge probe, from `results/p2_folds_partial.csv`: per-gene medians (M1 ridge, M2 smoothed), the pathway-level median that carries the headline result, and the immune-composition AUC.  $\lambda$  and  $\alpha$  are chosen by inner cross-validation on training patients only.

## R G2-X v2 per-pathway external transfer

Table 22: Phase-2 per-LOPO-fold medians.

| patient | $\lambda$ | $\alpha$ | M1 gene $r$ | M2 gene $r$ | pathway $r$ | immune AUC |
|---------|-----------|----------|-------------|-------------|-------------|------------|
| BC23287 | 100       | 0.5      | 0.059       | 0.072       | 0.642       | 0.783      |
| BC23450 | 100       | 0.5      | 0.039       | 0.042       | 0.246       | 0.573      |
| BC23567 | 100       | 0.5      | 0.033       | 0.039       | 0.104       | 0.520      |
| BC23810 | 100       | 0.5      | 0.062       | 0.068       | 0.101       | 0.577      |
| BC23901 | 100       | 0.5      | 0.042       | 0.038       | 0.400       | 0.677      |
| BC23903 | 100       | 0.5      | 0.106       | 0.117       | 0.614       | 0.743      |
| BC23944 | 100       | 0.5      | 0.039       | 0.044       | 0.252       | 0.590      |
| BC24044 | 100       | 0.5      | 0.085       | 0.098       | 0.571       | 0.793      |
| BC24105 | 100       | 0.5      | 0.031       | 0.038       | 0.093       | 0.577      |
| BC24220 | 100       | 0.5      | 0.051       | 0.061       | 0.497       | 0.696      |

All 50 pathway-level Pearson correlations behind the locked G2-X v2 verdict, from `results/g2x_external_eval_v2.json` (4,325 external 10x Visium spots, frozen head, Reinhard-normalized tiles; gate median  $\geq 0.10$  FAILED at  $-0.0728$ ). Sorted descending so the transfer gradient is inspectable: even the best-transferred pathway stays far below the gate.

| pathway                           | $r$    |
|-----------------------------------|--------|
| Angiogenesis                      | +0.226 |
| Apical Surface                    | +0.183 |
| Epithelial Mesenchymal Transition | +0.138 |
| Bile Acid Metabolism              | +0.105 |
| Glycolysis                        | +0.090 |
| Spermatogenesis                   | +0.083 |
| TGF-beta Signaling                | +0.082 |
| Coagulation                       | +0.064 |
| KRAS Signaling Up                 | +0.029 |
| Wnt-beta Catenin Signaling        | +0.023 |
| Myogenesis                        | +0.011 |
| heme Metabolism                   | +0.002 |
| Protein Secretion                 | -0.002 |
| Pancreas Beta Cells               | -0.002 |
| Hypoxia                           | -0.014 |
| UV Response Dn                    | -0.016 |
| E2F Targets                       | -0.018 |
| Inflammatory Response             | -0.026 |
| Myc Targets V1                    | -0.041 |
| Complement                        | -0.053 |
| IL-2/STAT5 Signaling              | -0.054 |
| Mitotic Spindle                   | -0.061 |
| Reactive Oxygen Species Pathway   | -0.062 |
| Myc Targets V2                    | -0.068 |
| mTORC1 Signaling                  | -0.072 |
| Cholesterol Homeostasis           | -0.074 |
| Apical Junction                   | -0.088 |
| KRAS Signaling Dn                 | -0.093 |

| pathway                       | <i>r</i> |
|-------------------------------|----------|
| Oxidative Phosphorylation     | -0.100   |
| Hedgehog Signaling            | -0.104   |
| Androgen Response             | -0.107   |
| Xenobiotic Metabolism         | -0.112   |
| IL-6/JAK/STAT3 Signaling      | -0.138   |
| UV Response Up                | -0.149   |
| p53 Pathway                   | -0.158   |
| TNF-alpha Signaling via NF-kB | -0.161   |
| Apoptosis                     | -0.172   |
| Peroxisome                    | -0.181   |
| Notch Signaling               | -0.183   |
| Fatty Acid Metabolism         | -0.183   |
| Allograft Rejection           | -0.191   |
| G2-M Checkpoint               | -0.194   |
| Unfolded Protein Response     | -0.195   |
| Estrogen Response Early       | -0.196   |
| Adipogenesis                  | -0.196   |
| DNA Repair                    | -0.267   |
| Interferon Gamma Response     | -0.317   |
| Interferon Alpha Response     | -0.326   |
| Estrogen Response Late        | -0.333   |
| PI3K/AKT/mTOR Signaling       | -0.382   |

Table 23: G2-X v2 per-pathway external-transfer correlations (sorted).

## S M3 per-fold results

Per-patient held-out medians for the M3 MLP nonlinear check against the committed M1/M2 values (same folds, same features; M3 spec in Table 1). M3 loses to the ridge in 9 of 10 folds.

Table 24: M3 vs M1/M2 per LOPO fold (median per-gene Pearson).

| patient | M1 ridge | M2 smoothed | M3 MLP  | M3 – M1 |
|---------|----------|-------------|---------|---------|
| BC23287 | 0.0598   | 0.0672      | 0.0047  | -0.0551 |
| BC23450 | 0.0427   | 0.0440      | 0.0177  | -0.0250 |
| BC23567 | 0.0214   | 0.0203      | 0.0196  | -0.0019 |
| BC23810 | 0.0286   | 0.0276      | 0.0320  | +0.0035 |
| BC23901 | 0.0093   | 0.0072      | -0.0115 | -0.0208 |
| BC23903 | 0.0686   | 0.0749      | 0.0522  | -0.0165 |
| BC23944 | 0.0338   | 0.0375      | 0.0059  | -0.0279 |
| BC24044 | 0.0710   | 0.0766      | 0.0612  | -0.0098 |
| BC24105 | -0.0016  | -0.0093     | -0.0172 | -0.0156 |
| BC24220 | 0.0458   | 0.0469      | 0.0257  | -0.0201 |

Table 25: Track B corrected-geometry rerun (results/trackB\_v2\_folds\_partial.csv): patient-held-out scanner performance on the corrected handcrafted features (features\_cache2, 100% of spots in-bounds on all 68 sections), identical recipe to the original run (class-balanced logistic regression and histogram gradient boosting, LOPO over the same 23 patients). Median AUC: LR 0.8915, GB 0.8772 - both above the locked G-B1 gate of 0.80, superseding the buggy-geometry measurements (0.642 / 0.676). The four folds below 0.70 (BC23270, BC24105, BC24223, BC23567) sit in near-degenerate or near-balanced label strata and are named, not removed.

| Patient | <i>n</i> spots | tumor frac | AUC (LR) | AUC (GB) | Brier (LR) |
|---------|----------------|------------|----------|----------|------------|
| BC23209 | 943            | 0.950      | 0.942    | 0.932    | 0.052      |
| BC23268 | 1425           | 0.182      | 0.795    | 0.813    | 0.134      |
| BC23269 | 1302           | 0.845      | 0.972    | 0.963    | 0.131      |
| BC23270 | 851            | 0.999      | 0.235    | 0.444    | 0.052      |
| BC23272 | 1691           | 0.671      | 0.875    | 0.877    | 0.285      |
| BC23277 | 1585           | 0.736      | 0.978    | 0.951    | 0.069      |
| BC23287 | 947            | 0.276      | 0.827    | 0.875    | 0.142      |
| BC23288 | 1422           | 0.907      | 0.808    | 0.767    | 0.135      |
| BC23377 | 1885           | 0.756      | 0.939    | 0.949    | 0.132      |
| BC23450 | 935            | 0.908      | 0.949    | 0.916    | 0.110      |
| BC23506 | 1473           | 0.940      | 0.964    | 0.973    | 0.069      |
| BC23508 | 1455           | 0.593      | 0.910    | 0.892    | 0.158      |
| BC23567 | 1729           | 0.485      | 0.689    | 0.757    | 0.428      |
| BC23803 | 1163           | 0.944      | 0.961    | 0.861    | 0.191      |
| BC23810 | 2099           | 0.881      | 0.697    | 0.773    | 0.116      |
| BC23895 | 1470           | 0.954      | 0.958    | 0.942    | 0.039      |
| BC23901 | 589            | 0.486      | 0.754    | 0.593    | 0.203      |
| BC23903 | 1373           | 0.199      | 0.892    | 0.878    | 0.106      |
| BC23944 | 1240           | 0.836      | 0.968    | 0.966    | 0.195      |
| BC24044 | 1728           | 0.123      | 0.918    | 0.932    | 0.113      |
| BC24105 | 960            | 0.999      | 0.693    | 0.531    | 0.043      |
| BC24220 | 1297           | 0.908      | 0.880    | 0.877    | 0.327      |
| BC24223 | 1093           | 0.091      | 0.656    | 0.667    | 0.158      |
| Median  |                |            | 0.8915   | 0.8772   |            |

Table 26: Track B corrected-geometry scanner: feature-level inference. statsmodels logistic fit on standardized corrected features, all sections; coefficients sorted by  $|z|$ . Stain and color channels (blue-channel mean, red-channel mean, eosin and DAB deconvolution means) carry the largest effects, with GLCM texture statistics behind them - the scanner reads stain composition and local disorganization, exactly the morphology a pathologist weighs. Supersedes the buggy-geometry inference (results/scanner\_coefficients.csv, preserved as audit trail).

| Feature         | Coef   | $z$    | $p$      |
|-----------------|--------|--------|----------|
| b_mean          | +9.659 | +17.30 | 4.54e-67 |
| r_mean          | -5.694 | -17.84 | 3.63e-71 |
| eosin_mean      | +5.144 | +12.06 | 1.69e-33 |
| dab_mean        | +4.933 | +13.69 | 1.24e-42 |
| glcm3_contrast  | +4.388 | +3.85  | 1.20e-04 |
| glcm1_energy    | -4.344 | -2.72  | 6.46e-03 |
| glcm3_corr      | +4.009 | +5.54  | 3.03e-08 |
| glcm1_contrast  | -3.606 | -3.32  | 9.15e-04 |
| glcm1_corr      | -3.581 | -4.95  | 7.29e-07 |
| r_kurt          | -3.573 | -5.68  | 1.32e-08 |
| glcm2_homog     | -3.199 | -5.48  | 4.21e-08 |
| hema_mean       | +2.649 | +9.74  | 1.94e-22 |
| glcm0_energy    | +2.168 | +1.38  | 1.68e-01 |
| r_std           | +2.140 | +11.38 | 5.33e-30 |
| b_skew          | +2.139 | +11.62 | 3.25e-31 |
| glcm0_homog     | +1.951 | +3.31  | 9.27e-04 |
| glcm2_contrast  | +1.940 | +1.68  | 9.34e-02 |
| b_kurt          | +1.906 | +6.72  | 1.76e-11 |
| hema_std        | -1.750 | -14.52 | 9.56e-48 |
| eosin_std       | +1.539 | +8.17  | 3.18e-16 |
| glcm2_corr      | +1.431 | +1.95  | 5.10e-02 |
| g_std           | -1.431 | -6.72  | 1.77e-11 |
| gray_entropy    | -1.385 | -16.11 | 2.15e-58 |
| glcm0_contrast  | -1.309 | -1.19  | 2.36e-01 |
| b_std           | -1.292 | -8.54  | 1.29e-17 |
| glcm0_corr      | -1.232 | -1.69  | 9.10e-02 |
| glcm2_energy    | +1.212 | +0.97  | 3.32e-01 |
| g_mean          | -1.150 | -3.38  | 7.21e-04 |
| hema_dense_frac | +0.952 | +3.44  | 5.86e-04 |
| eosin_fracpos   | -0.764 | -11.22 | 3.33e-29 |
| glcm3_energy    | +0.731 | +0.57  | 5.66e-01 |
| glcm1_homog     | +0.603 | +1.03  | 3.03e-01 |
| g_kurt          | -0.541 | -1.20  | 2.29e-01 |
| glcm3_homog     | -0.538 | -0.94  | 3.48e-01 |

| Feature       | Coef   | $z$   | $p$      |
|---------------|--------|-------|----------|
| hema_fracpos  | -0.435 | -5.39 | 6.97e-08 |
| grad_mag_mean | +0.397 | +2.19 | 2.82e-02 |
| g_skew        | -0.300 | -1.32 | 1.87e-01 |
| hema_entropy  | +0.205 | +5.09 | 3.52e-07 |
| r_skew        | +0.155 | +0.66 | 5.08e-01 |
| dab_fracpos   | +0.131 | +2.72 | 6.50e-03 |
| dab_std       | +0.129 | +2.00 | 4.52e-02 |

Table 27: Learned versus handcrafted representation, per held-out patient. Handcrafted: Track A M1 ridge on 41 morphology features (corrected-geometry rerun); learned: Phase-2 ridge on 768-dimensional CTransPath embeddings. Values are median per-gene Pearson  $r$  on the held-out patient’s spots. The learned representation wins 8 of 10 folds (median paired delta +0.0131, Wilcoxon  $p = 0.0098$ ); both medians remain far below the locked G2-R gate of 0.08.

| Patient | Handcrafted $r$ | Learned $r$ | $\Delta$ (learned – handcrafted) |
|---------|-----------------|-------------|----------------------------------|
| BC23287 | 0.0598          | 0.0588      | -0.0010                          |
| BC23450 | 0.0427          | 0.0388      | -0.0039                          |
| BC23567 | 0.0214          | 0.0334      | +0.0120                          |
| BC23810 | 0.0286          | 0.0623      | +0.0337                          |
| BC23901 | 0.0093          | 0.0417      | +0.0324                          |
| BC23903 | 0.0686          | 0.1063      | +0.0377                          |
| BC23944 | 0.0338          | 0.0390      | +0.0052                          |
| BC24044 | 0.0710          | 0.0853      | +0.0142                          |
| BC24105 | -0.0016         | 0.0312      | +0.0328                          |
| BC24220 | 0.0458          | 0.0507      | +0.0049                          |
| Median  | 0.0382          | 0.0462      | +0.0131                          |

Table 28: Nested evaluation of the 250-gene selection rule (pre-declared rules, queue #11). Per-fold median per-gene Pearson  $r$  on the held-out patient for four selection rules: top-100 / top-250 / top-500 by train-fold mean expression, and top-250 by train-fold variance. The last column is the gene overlap between the mean- and variance-selected 250-gene panels. The headline median moves by at most  $\pm 0.005$  across rules and every rule remains far below the locked G2-R gate of 0.08 - the gate conclusion is selection-rule-robust.

| Patient | top-100 mean | top-250 mean | top-500 mean | top-250 var | mean $\cap$ var |
|---------|--------------|--------------|--------------|-------------|-----------------|
| BC23287 | 0.0366       | 0.0520       | 0.0775       | 0.0767      | 155             |

| Patient | top-100 mean | top-250 mean | top-500 mean | top-250 var | mean $\cap$ var |
|---------|--------------|--------------|--------------|-------------|-----------------|
| BC23450 | 0.0836       | 0.0484       | 0.0473       | 0.0545      | 172             |
| BC23567 | 0.0420       | 0.0404       | 0.0357       | 0.0392      | 168             |
| BC23810 | 0.1121       | 0.0720       | 0.0471       | 0.0592      | 176             |
| BC23901 | 0.0402       | 0.0314       | 0.0353       | 0.0303      | 172             |
| BC23903 | 0.1197       | 0.1006       | 0.0894       | 0.0958      | 171             |
| BC23944 | 0.0338       | 0.0323       | 0.0321       | 0.0326      | 177             |
| BC24044 | 0.0765       | 0.0846       | 0.0889       | 0.0855      | 161             |
| BC24105 | 0.0509       | 0.0315       | 0.0251       | 0.0356      | 170             |
| BC24220 | 0.0633       | 0.0542       | 0.0492       | 0.0547      | 174             |
| Median  | 0.0571       | 0.0502       | 0.0472       | 0.0546      | 171.5           |

Table 29: Gene-level negative control (pre-declared, queue #12). Control column: median across 5 pre-declared draws of 250 genes from train-expression ranks 251-2250 (seed 20260929), same ridge harness. Selected-250: the committed ridge on the fold-varying top-250 panel. PLS: the multi-task winner. The control's pre-declared  $\approx 0$  expectation was falsified - per-gene predictability is broad across expressed genes (cell-type-composition-driven), not panel-specific - but the selected panel beats controls in 9/10 folds (Wilcoxon  $p = 0.0488$ ) and PLS in 10/10 ( $p = 0.0020$ ). Named anomaly: BC23287, controls  $0.0814 > \text{selected } 0.0588$ .

| Patient | Control median $r$ | Selected-250 ridge $r$ | PLS $r$ |
|---------|--------------------|------------------------|---------|
| BC23287 | 0.0806             | 0.0588                 | 0.0896  |
| BC23450 | 0.0303             | 0.0388                 | 0.0734  |
| BC23567 | 0.0223             | 0.0334                 | 0.0491  |
| BC23810 | 0.0220             | 0.0623                 | 0.0786  |
| BC23901 | 0.0365             | 0.0417                 | 0.0494  |
| BC23903 | 0.0654             | 0.1063                 | 0.1077  |
| BC23944 | 0.0298             | 0.0390                 | 0.0442  |
| BC24044 | 0.0865             | 0.0853                 | 0.0966  |
| BC24105 | 0.0112             | 0.0312                 | 0.0400  |
| BC24220 | 0.0390             | 0.0507                 | 0.0611  |
| Median  | 0.0334             | 0.0462                 | 0.0673  |

Table 30: Spatial structure of the expression task, per held-out patient. kNN / Gaussian oracles: median per-gene  $r$  of smoothing predictors given the *true* expression at neighboring held-out spots (an upper bound on what pure spatial autocorrelation carries). Moran’s  $I$ : median spatial autocorrelation of the 250-gene panel; its correlation with the kNN-oracle  $r$  shows per-gene predictability tracks autocorrelation. Gain: M2 (spatial-feature model) minus M1 held-out  $r$ ; shell p95: 95th percentile of the same gain under distance-shell permutations (the strictest of three pre-declared nulls).

| Patient | kNN oracle | Gaussian oracle | Moran’s $I$ | corr( $I$ , kNN $r$ ) | M2–M1 gain | shell null p95 |
|---------|------------|-----------------|-------------|-----------------------|------------|----------------|
| BC23287 | 0.123      | 0.146           | 0.029       | 0.89                  | +0.0138    | 0.0173         |
| BC23450 | 0.110      | 0.137           | 0.038       | 0.94                  | -0.0001    | 0.0070         |
| BC23567 | 0.166      | 0.214           | 0.071       | 0.98                  | +0.0074    | 0.0099         |
| BC23810 | 0.112      | 0.151           | 0.045       | 0.96                  | +0.0053    | 0.0024         |
| BC23901 | 0.114      | 0.141           | 0.038       | 0.95                  | -0.0041    | 0.0033         |
| BC23903 | 0.158      | 0.184           | 0.053       | 0.96                  | +0.0114    | 0.0096         |
| BC23944 | 0.108      | 0.143           | 0.042       | 0.95                  | +0.0075    | 0.0165         |
| BC24044 | 0.165      | 0.201           | 0.062       | 0.97                  | +0.0133    | 0.0095         |
| BC24105 | 0.105      | 0.116           | 0.034       | 0.96                  | +0.0067    | 0.0072         |
| BC24220 | 0.090      | 0.114           | 0.033       | 0.97                  | +0.0071    | 0.0091         |
| Median  | 0.113      | 0.144           | 0.040       | 0.96                  | +0.0073    | 0.0093         |

## T Scanner calibration reliability table

The full 10-bin reliability table behind the calibration diagnostics, from `results/p2_scanner_calibration.json` (12,897 out-of-fold spots; Brier 0.172, prevalence baseline 0.240, ECE 0.141).

Table 31: Scanner reliability by predicted-probability bin.

| bin     | $n$   | mean predicted | empirical tumor rate | gap    |
|---------|-------|----------------|----------------------|--------|
| 0.0–0.1 | 3,588 | 0.015          | 0.140                | -0.124 |
| 0.1–0.2 | 471   | 0.146          | 0.433                | -0.288 |
| 0.2–0.3 | 372   | 0.249          | 0.578                | -0.329 |
| 0.3–0.4 | 322   | 0.350          | 0.643                | -0.293 |
| 0.4–0.5 | 327   | 0.449          | 0.667                | -0.217 |
| 0.5–0.6 | 335   | 0.547          | 0.716                | -0.170 |
| 0.6–0.7 | 441   | 0.650          | 0.755                | -0.106 |
| 0.7–0.8 | 536   | 0.753          | 0.750                | +0.003 |
| 0.8–0.9 | 855   | 0.856          | 0.794                | +0.062 |
| 0.9–1.0 | 5,650 | 0.979          | 0.840                | +0.140 |

## U Transport cohort, all 39 sections

Every external-transport measurement behind G-A4, from `results/trackA_results.json`: the HER2+-trained M1 ridge applied to Luminal A, Luminal B, and TNBC sections with no refit. Sorted by subtype

then correlation; the degradation versus in-subtype held-out performance is the expected, honestly-logged negative.

| section                               | subtype   | median $r$ |
|---------------------------------------|-----------|------------|
| <b>Luminal_A</b> (n=12, median 0.070) |           |            |
| BC23268_D1                            | Luminal_A | 0.1311     |
| BC23268_C2                            | Luminal_A | 0.1124     |
| BC23268_C1                            | Luminal_A | 0.1122     |
| BC24223_E1                            | Luminal_A | 0.0872     |
| BC24223_D2                            | Luminal_A | 0.0864     |
| BC23272_E1                            | Luminal_A | 0.0697     |
| BC23272_D2                            | Luminal_A | 0.0565     |
| BC23269_C1                            | Luminal_A | 0.0450     |
| BC24223_E2                            | Luminal_A | 0.0438     |
| BC23272_E2                            | Luminal_A | 0.0374     |
| BC23269_D1                            | Luminal_A | 0.0318     |
| BC23269_C2                            | Luminal_A | 0.0186     |
| <b>Luminal_B</b> (n=15, median 0.027) |           |            |
| BC23508_D2                            | Luminal_B | 0.0578     |
| BC23277_E1                            | Luminal_B | 0.0557     |
| BC23270_E1                            | Luminal_B | 0.0505     |
| BC23508_E1                            | Luminal_B | 0.0453     |
| BC23508_E2                            | Luminal_B | 0.0373     |
| BC23277_D2                            | Luminal_B | 0.0323     |
| BC23895_C1                            | Luminal_B | 0.0288     |
| BC23895_D1                            | Luminal_B | 0.0273     |
| BC23506_C1                            | Luminal_B | 0.0244     |
| BC23506_C2                            | Luminal_B | 0.0026     |
| BC23895_C2                            | Luminal_B | 0.0007     |
| BC23270_D2                            | Luminal_B | -0.0032    |
| BC23270_E2                            | Luminal_B | -0.0032    |
| BC23506_D1                            | Luminal_B | -0.0045    |
| BC23277_E2                            | Luminal_B | -0.0133    |
| <b>TNBC</b> (n=12, median 0.029)      |           |            |
| BC23209_D1                            | TNBC      | 0.0711     |
| BC23209_C2                            | TNBC      | 0.0553     |
| BC23288_E2                            | TNBC      | 0.0536     |
| BC23288_E1                            | TNBC      | 0.0437     |
| BC23803_E1                            | TNBC      | 0.0385     |
| BC23288_D2                            | TNBC      | 0.0286     |
| BC23209_C1                            | TNBC      | 0.0188     |
| BC23803_D2                            | TNBC      | 0.0049     |
| BC23377_C1                            | TNBC      | -0.0088    |
| BC23803_E2                            | TNBC      | -0.0186    |
| BC23377_D1                            | TNBC      | -0.0277    |
| BC23377_C2                            | TNBC      | -0.0348    |

| section | subtype | median $r$ |
|---------|---------|------------|
|---------|---------|------------|

Table 32: G-A4 transport: per-section held-out median per-gene Pearson on non-HER2 subtypes.

## V Per-pathway niche spatial architecture (D2)

All 50 pathway-niche Moran's  $I$  values behind the locked D2 result, from `results/p2_niches_summary.json` ( $k = 8$  spatial niches per section, 200 null relabelings; 29 HER2 sections). Sorted by median  $I$  across sections;  $\text{frac } p < 0.05$  is the fraction of sections where the niche's pathway score is significantly spatially clustered. Cohort medians: niche  $I$  0.213; 96.6% of sections have all niches significant; the PancreasBeta control sits at 0.072.

| pathway niche                     | median $I$ | frac $p < 0.05$ |
|-----------------------------------|------------|-----------------|
| Myc Targets V1                    | 0.362      | 1.00            |
| Mitotic Spindle                   | 0.346      | 1.00            |
| G2-M Checkpoint                   | 0.340      | 1.00            |
| mTORC1 Signaling                  | 0.339      | 1.00            |
| E2F Targets                       | 0.329      | 1.00            |
| Glycolysis                        | 0.308      | 1.00            |
| Epithelial Mesenchymal Transition | 0.305      | 1.00            |
| Androgen Response                 | 0.303      | 1.00            |
| Oxidative Phosphorylation         | 0.302      | 1.00            |
| Unfolded Protein Response         | 0.301      | 1.00            |
| Interferon Gamma Response         | 0.288      | 1.00            |
| Xenobiotic Metabolism             | 0.287      | 1.00            |
| Protein Secretion                 | 0.285      | 1.00            |
| Fatty Acid Metabolism             | 0.278      | 1.00            |
| Hypoxia                           | 0.277      | 1.00            |
| Adipogenesis                      | 0.275      | 1.00            |
| DNA Repair                        | 0.274      | 1.00            |
| Apical Junction                   | 0.273      | 1.00            |
| Estrogen Response Early           | 0.270      | 1.00            |
| Cholesterol Homeostasis           | 0.265      | 1.00            |
| TNF-alpha Signaling via NF-kB     | 0.262      | 1.00            |
| heme Metabolism                   | 0.259      | 1.00            |
| UV Response Up                    | 0.259      | 1.00            |
| Estrogen Response Late            | 0.258      | 1.00            |
| Pperoxisome                       | 0.256      | 1.00            |
| p53 Pathway                       | 0.255      | 0.97            |
| UV Response Dn                    | 0.255      | 1.00            |
| Complement                        | 0.252      | 1.00            |
| Apoptosis                         | 0.247      | 0.97            |
| PI3K/AKT/mTOR Signaling           | 0.247      | 1.00            |
| IL-2/STAT5 Signaling              | 0.238      | 1.00            |
| Interferon Alpha Response         | 0.232      | 1.00            |

| pathway niche                   | median $I$ | frac $p < 0.05$ |
|---------------------------------|------------|-----------------|
| Myogenesis                      | 0.231      | 1.00            |
| Coagulation                     | 0.229      | 1.00            |
| KRAS Signaling Up               | 0.222      | 0.93            |
| Allograft Rejection             | 0.202      | 1.00            |
| TGF-beta Signaling              | 0.199      | 0.93            |
| Inflammatory Response           | 0.195      | 0.97            |
| Angiogenesis                    | 0.184      | 0.90            |
| Myc Targets V2                  | 0.183      | 0.93            |
| Bile Acid Metabolism            | 0.165      | 0.93            |
| Reactive Oxygen Species Pathway | 0.164      | 0.97            |
| Spermatogenesis                 | 0.163      | 0.93            |
| IL-6/JAK/STAT3 Signaling        | 0.144      | 0.97            |
| Notch Signaling                 | 0.142      | 0.90            |
| Apical Surface                  | 0.137      | 0.93            |
| Wnt-beta Catenin Signaling      | 0.125      | 0.83            |
| KRAS Signaling Dn               | 0.114      | 0.90            |
| Hedgehog Signaling              | 0.105      | 0.83            |
| Pancreas Beta Cells             | 0.072      | 0.83            |

Table 33: Per-pathway niche spatial clustering (sorted by median Moran's  $I$ ).